# Transfer Learning of Spatial Features From High-Resolution RGB Images for Large-Scale and Robust Hyperspectral Remote Sensing Target Detection

Yuanfeng Wu, *Senior Member, IEEE*, Zijin Li, Boya Zhao, Yuhang Song, and Bing Zhang, *Fellow, IEEE*

***Abstract*— Target detection is a critical task in interpreting hyperspectral remote sensing images. Small target (such as airplanes) detection is challenging, especially in large-scale complex scenes with high spectral variability of different land cover types. In this article, we propose a transfer learning-based, large-scale, robust hyperspectral target detector (TLH$^2$TD) to improve the accuracy of hyperspectral target detection (HTD) in large-scale complex scenes. TLH$^2$TD learns the spatial features of hyperspectral targets from high-resolution remote sensing images and achieves high-precision HTD with fused spatial–spectral features. It comprises three parts. First, the coupled target-background sample expansion (CTBSE) module is designed to expand the labeled hyperspectral target and background samples with sufficient high-resolution, labeled RGB images, and a few labeled hyperspectral samples. Second, the hard positive and negative example mining (HPNEM) module trains the hard positive and negative samples to enhance the discriminative ability of the network, addressing the problem of inadequate sample training in large-scale hyperspectral images (HSIs). Third, the spatial–spectral weighted subspace (SSWS) module is designed to fuse the spatial features extracted from the target detection network and the spectral features based on the Mahalanobis distance. The results show that: 1) the TLH$^2$TD achieves average area under the curve (AUC) values of 0.96, 0.96, and 0.93 on small-, medium-, and large-sized HSIs, respectively, achieving the highest accuracy compared with other HTD algorithms; 2) the TLH$^2$TD exhibits the highest detection time efficiency for medium- and large-sized HSIs; and 3) for large-sized HSIs, when most HTD algorithms fail, TLH$^2$TD exhibits significantly higher accuracy and time efficiency than other methods. This is an important achievement in meeting the robust target detection tasks of large-scale hyperspectral remote sensing images.**

***Index Terms*— Hyperspectral images (HSIs), hyperspectral target detection (HTD), large-scale complex scenes, spatial–spectral feature fusion, transfer learning.**

Manuscript received 30 October 2023; revised 2 January 2024; accepted 14 January 2024. Date of publication 17 January 2024; date of current version 31 January 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203 and in part by the National Natural Science Foundation of China under Grant 62001455 and Grant 41871245. *(Corresponding author: Boya Zhao.)*

Yuanfeng Wu and Boya Zhao are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: wuyf@aircas.ac.cn; zhaoby@aircas.ac.cn).

Zijin Li is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: lizijin21@mails.ucas.ac.cn).

Yuhang Song and Bing Zhang are with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: songyuhang23@mails.ucas.ac.cn; zb@radi.ac.cn).

Digital Object Identifier 10.1109/TGRS.2024.3355184

## I. Introduction

HYPERSPECTRAL remote sensing refers to the combination of imaging and spectroscopy in a system using high spectral resolution to obtain 2-D spatial features and 1-D spectral information of target areas [1], [2]. Each pixel in a hyperspectral image (HSI) contains spectral information in dozens or hundreds of bands in the visible to short-wave infrared range (400–2500 nm), providing high spectral resolution target information with the potential for detailed feature classification and material identification. Hyperspectral target detection (HTD) is based on the differences in spectral features between the target and the background; thus, targets that are difficult to identify based on morphological features or are not easily detected using multispectral imagery can be detected in HSIs [3].

High spatial and spectral resolutions cannot be obtained simultaneously due to the constraints of the image sensor [4]. Unlike high-resolution visible-band satellite sensors, hyperspectral imaging sensors usually have lower spatial resolution due to a large number of wavelength bands. With the rapid development of space remote sensing technology, the spatial resolution of hyperspectral satellite sensors has been significantly improved [5]. For onboard hyperspectral satellites, the wide swath and the large number of spectral bands lead to a huge data volume of HSIs, making it necessary to design more efficient algorithms for large-scale HSIs. As shown in Table I, the spatial resolution of China's Zhuhai-1 satellite sensor and the SHALOM satellite sensor, jointly developed by Italy and Israel, is 10 m. The spatial resolution of hyperspectral sensors will improve in the future and reach meter-level resolution. The spatial characteristics of targets will become discernable in HSIs, enabling high-precision target detection. As shown in Fig. 1, in Hyperion or Gaofen-5 HSIs with a spatial resolution of 30 m, the airport can be easily detected, while an aircraft on the tarmac covers only about 2–5 pixels (shown as a few bright spots in the HSIs). In the HSI with a spatial resolution

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

of 3.5 m, an aircraft covers 20–100 pixels, providing more morphological information.

HTD assumes that the target and background obey a mathematical model, and target detection is carried out using spectral statistics. HTD can be categorized as anomaly detection (AED) and matched detection, depending on the presence/absence of prior knowledge of the target [6].

AED algorithms are unsupervised algorithms because there is no prior knowledge of the target [7]. They utilize multivariate normal distribution models and probabilistic statistics, such as the Reed–Xiaoli detector (RXD) algorithm [8], collaborative representation-based AED (CRD) [9], uniform target detector (UTD) [10], kernel-RX [11], the local adaptive iterative RX algorithm [12], Deriv-RX [13], and the fractional Fourier entropy (FrFE)-based RX [14].

Matched detection algorithms include the adaptive matched filter (AMF) [15], adaptive coherence estimator (ACE) [16], constrained energy minimization (CEM) [17], and target-constrained interference minimized filter (TCIMF) based on CEM [18].

Methods that consider both spatial and spectral features have been used recently for HTD with the improvement of spatial resolution of HSIs, such as the attribute and edge-preserving filtering-based AED [19], matched subspace detector (MSD) [20], spectral–spatial stacked autoencoders based on a bilateral filter (SSSAE-BF) [21], and the total variation regularized low-rank tensor decomposition and collaborative representation detector (LRTDCRD) [22]. These methods utilize several approaches to extract morphological features to facilitate target detection, including filters, tensor decomposition, and simple morphological operations, without exploiting deeper features.

In addition, deep learning approaches have been widely used in HTD. Due to the lack of sufficient samples for network training, relatively few deep-learning methods for HTD are based on supervised approaches. For example, Du et al. [23] proposed an HTD algorithm based on a convolutional neural network (CNNTD), which expands labels by subtracting target and background pixels to construct an adequate training set but requires manually labeling a large number of additional target and background labels. In [24], the pixel pairs generated from the reference image scene were used to expand the samples and as the classification dataset to train multilayer CNNs; however, the training results depend on the completeness of the reference image scene. HTD-Net proposed by Zhang et al. [25] utilizes an autoencoder and a linear prediction strategy to construct target-background pixel pairs to expand the training set, but the process is relatively complicated and lacks rationality.

Unlike supervised approaches, unsupervised, self-supervised, and other methods do not require large amounts of labeled data. For example, the deep spatial–spectral network (DSSN) proposed by Shi et al. [26] performs HTD in an unsupervised manner. Other studies used self-supervised methods, such as the memory-augmented autoencoder for hyperspectral AED (MAENet) [27], the graph regularized variational autoencoder (GRVAE) [28], the local spatial–spectral information-integrated semisupervised two-stream network (LS$^3$T-Net) [29], and the blind spot self-supervised learning network (BS$^3$LNet) [30]. These algorithms detect targets based on the characteristics of the sparse distribution and small target size in HSIs and utilize autoencoders to reconstruct the background; they often fail to reconstruct the target. These HTD algorithms are usually trained on a single or very few HSIs and are prone to overfitting on account of insufficient training data or noisy features.

In summary, current HTD algorithms for large-scale HSIs with complex scenes have three challenges. First, spectral feature-based algorithms have a high false alarm rate and low time efficiency for large-scale HSIs with complex scenes. As shown in Fig. 2, the spectral features of bright and dark airplanes vary substantially, and the spectral signatures of background pixels, such as runways and roofs, are difficult to distinguish from dark airplanes. Second, existing spatial feature-based HTD algorithms mostly rely on low-level morphological features, making it difficult to distinguish targets and background. Third, the lack of public labeled HSI datasets restricts the development of HTD based on deep learning. Therefore, a high-precision algorithm for large-scale and robust HTD is urgently needed.

To address the above problems, this article proposes a transfer learning-based, large-scale, robust hyperspectral target detector (TLH$^2$TD) to achieve high-precision large-scale HTD with insufficient target samples and reduce the false alarm rate. The main contributions of this article are summarized as follows.

1) To address insufficient target samples and weak characterization ability, we propose a novel coupled target-background sample expansion (CTBSE) method to transfer spatial features from high spatial resolution images containing similar small targets for HTD. Unlike the strategy of expanding target samples using autoencoders or typical background pixels, the proposed method accurately simulates the spatial morphological information of target pixels in HSIs while maximally preserving the a priori target spectral information.
2) We propose a mining mechanism for difficult-to-distinguish positive and negative samples to address the insufficient characterization of positive samples and insufficient training in large-scale HSIs. This strategy balances the ratio between target and background samples, enabling the network to learn more about difficult-to-distinguish positive and negative samples at the boundary of the classification threshold. This method solves the problem of low sample sizes for training in large-scale HSIs and improves the accuracy of target detection.
3) We propose a spatial–spectral feature-associated HSI target detection method based on the spatial–spectral weighted subspace (SSWS) to address the high spectral variability of features in complex hyperspectral scenes and enable the model to distinguish the target from the background using fused features. This method utilizes the spectral features and depth convolution features in HSIs, projecting the features into a high-dimensional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS OF CHINESE AND INTERNATIONAL HYPERSPECTRAL REMOTE SENSING SATELLITE SENSORS IN RECENT YEARS

| Satellites/ Sensors | Country | Launch time | Bands | Wavelength (nm) | Spatial Resolution (m) | Spectral Resolution (nm) | Swath width (km) | Number of pixels covered by an aircraft (30mx30m) |
|---|---|---|---|---|---|---|---|---|
| Gaofen-5 | China | 2018 | 330 | 400-2500 | 30 | 5(VNIR) 10(SWIR) | 60 | 1 |
| Zhuhai-1 OHS | China | 2018 | 32 | 400-1000 | 10 | 2.5 | 160 | 9 |
| HySIS | India | 2018 | 326 | 400-2500 | 30 | 10 | 30 | 1 |
| PRISMA | Italy | 2019 | 239 | 400-2500 | 30 | $\leq$12 | 30 | 1 |
| ZY-1 02D | China | 2019 | 166 | 400-2500 | 30 | 10(VNIR) 20(SWIR) | 60 | 1 |
| DESIS | Germany | 2019 | 235 | 400-1000 | 30 | 3.3 | 24.7$\sim$33.4 | 1 |
| CartoSat-3 | India | 2019 | 200 | 400-2400 | 12 | / | 5 | 6.25 |
| EnMAP | Germany | 2020 | 228 | 420-2450 | 30 | 6.5(VNIR) 10(SWIR) | 30 | 1 |
| ZY-1 02E | China | 2021 | 166 | 400-2500 | 30 | 10(VNIR) 20(SWIR) | 60 | 1 |
| ALOS-3 | Japan | 2023 | 185 | 400-2500 | 30 | 10(VNIR) 12.5(SWIR) | 30 | 1 |
| SHALOM | Italy/Israel | To be launched (2025) | 241 | 400-2500 | 10 | 10 | 10 | 9 |

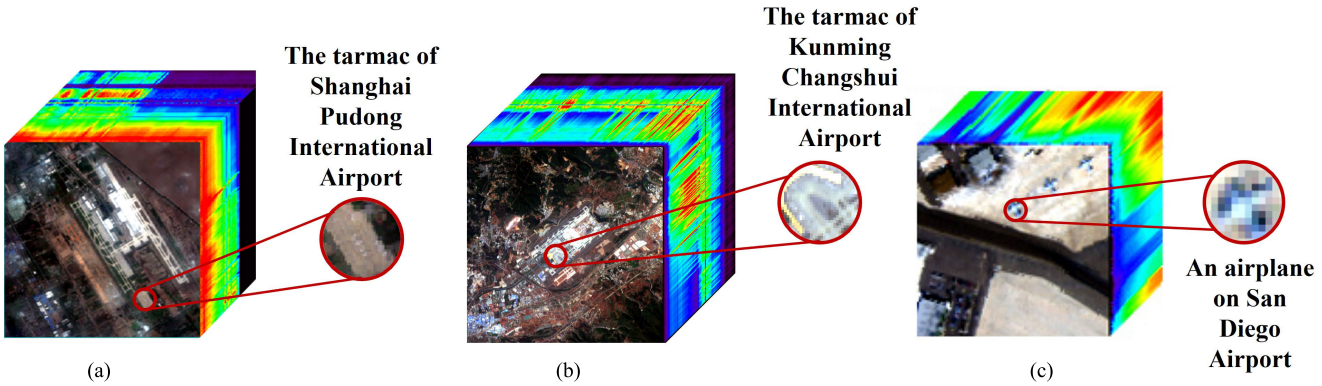



Fig. 1. Relationship between morphological information of targets and spatial resolution of HSIs. (a) Hyperion, spatial resolution 30 m. (b) Gaofen-5, spatial resolution 30 m. (c) AVIRIS, spatial resolution 30 m.

space to improve the target/background separability. The proposed method accurately reflects the spatial–spectral characteristics of HSIs and improves the target detection accuracy.

The remainder of this article is organized as follows. Section II describes the related work in HTD. Section III details the proposed HTD algorithm. Section IV presents and analyzes the experimental results, including a comparison with existing state-of-the-art methods. Section V concludes this article.

## II. RELATED WORK

### A. Transfer Learning for Hyperspectral Data

Transfer learning aims to leverage existing data, models, and knowledge to apply models from one domain to another domain, scenario, and task through domain similarity, reducing the model's dependence on massive amounts of labels and ensuring more powerful generalization capabilities of machine learning methods [31], [32], [33]. HSIs contain spatial and spectral information on the target. The spatial features are similar to those of visible band remote sensing images; only the resolution differs. Labeled data from visible band imagery and a small amount of labeled hyperspectral data can be utilized to perform transfer learning to characterize the features.

Scholars have proposed transfer learning for HSI processing using a small amount of labeled data in recent years. For example, Jun and Ghosh [34] proposed an HSI classification algorithm based on knowledge transfer, achieving efficient learning of transfer knowledge by adjusting the distribution of the labeled dataset. Du et al. [35] proposed an unsupervised

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

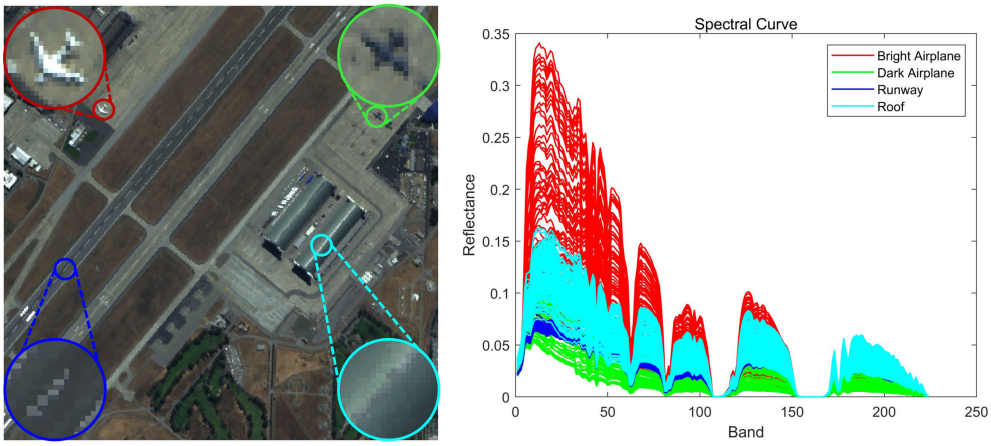



Fig. 2. Spectral curves of different objects: bright airplane (red), dark airplane (green), runway (blue), and roof (cyan).

transfer learning-based target detection (UTLD) method for HTD. It uses multivariate outlier analysis and segmentation to obtain labeled and unlabeled positive and negative samples to construct a target detection subspace based on transfer learning. Shen et al. [36] proposed a hyperspectral feature adaptation and augmentation (HFAA) method that learns a common subspace by utilizing two projection matrices to determine the features of the source and target domains and improve the similarity of the samples with the same labels in both domains for cross-scene hyperspectral classification. Bao and Zhu [37] mapped the feature space of the source dataset to the target dataset for HTD based on feature transfer learning. Li et al. [38] proposed a novel iterative reweighting heterogeneous transfer learning (IRHTL) framework for HSI classification. It uses source and target data mapped to a common space through iterative learning and a new iterative weighting strategy to adjust the source data sample weights. Shi et al. [39] proposed a semisupervised domain adaptive few-shot learning (SDAFL) model that transfers the knowledge learned from the large-sample target domain to the small-sample target domain. Lei et al. [40] proposed a spatial–spectral associated detector (SSAD) based on a semantic transferred priori approach for HTD. It uses a transfer learning method for the feature alignment of the source and target domain data and extracts a customized and accurate target spectrum from the target spatial domain using a semantic segmentation network. Subsequently, the CEM algorithm is used for spectral dimensionality detection and to obtain the final result.

Existing HTD methods based on transfer learning only focus on the transformation of target samples but do not consider the equalization of target and background samples and the transformation of background samples in large-scale HSIs. In this article, the target and background in the visible band and HSI data are feature-aligned based on CTBSE and hard example mining to enhance the separability of target and background samples in large-scale HSIs.

### *B. Joint Spatial–Spectral Feature-Based HTD*

The improvement in the spatial resolution of HSIs enables HTD to use spatial and spectral features. Feature fusion methods, such as the attribute and edge-preserving filtering-based hyperspectral AED method proposed by Kang et al. [19], obtain initial pixel detection results using spectral features and area attributes. Subsequently, the initial detection results are improved with edge-preserving filtering, exploiting the spatial correlation between neighboring pixels. The tensor MSD proposed by Liu et al. [20] utilizes third-order tensor blocks to represent hyperspectral targets and backgrounds. It projects the tensor blocks into the tensor subspace and detects targets based on the ratio of residual energies. The SSSAE-BF [21] uses bilateral filters to obtain anomalous and background components and uses a stacked autoencoder (SAE) to obtain the spatial–spectral feature matrix. It then uses the RX detector on the spatial–spectral feature matrix to obtain the final detection results. The LRTDCRD [22] uses the total variation regularized low-rank tensor decomposition (LRTDTV) model to divide the HSI into two parts (the low-rank background and mixed information) and detects the targets in the latter part using a domain-transformed recursive filter (DTRF). Subsequently, it applies the CRD to extract the anomalous target embeddings in the low-rank part and obtains the final detection results by fusing the initial results of the low-rank background and the mixed information.

Decision fusion methods, such as the LS$^3$T-Net proposed by Wang et al. [29], extract spatial and spectral features separately using a two-stream network and fuse the detection results. Huang et al. [41] proposed a game theory-based AED method for HSIs. The spectral features, extended morphological features (EMPs), and Gabor texture features of HSIs are acquired, and three initial detections are generated based on the superpixels of different features. Saliency-based decision fusion combines the complementary information in different features to obtain a fused detection map. Tu et al. [42] proposed a hyperspectral AED method based on quantum potential clustering. It first obtains a saliency distribution map using an unsupervised saliency strategy and then obtains an initial detection map using wave function maps and quantum potential clustering. The saliency map is fused with the initial detection map to obtain the final detection result. Song et al. [43] proposed a spectral–spatial anomaly hyperspectral detector based on an improved isolation forest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

(SSIIFD) that utilizes the Gabor filter and the entropy rate segmentation (ERS) algorithm to extract spatial and spectral features, respectively. Linear combinations of the spatial and spectral anomaly scores are created to predict anomalous pixels.

## III. Method

Fig. 3 shows the framework of the proposed TLH$^2$TD, which comprises three parts. First, the CTBSE module enhances the diversity of the hyperspectral target and background samples by combining several data augmentation methods. Second, the hard positive and negative example mining (HPNEM) module enables the network to learn the HSI target and background spatial representations and balance the relationship between positive and negative samples to achieve transfer learning-based network training. Finally, the SSWS module distinguishes targets and the background in a higher dimensional fused feature subspace for large-scale HTD. Some missing targets and false alarms may occur in the spatial or spectral feature space (e.g., the aircraft with the spatial feature $\mathbf{V}(r) = 0.39$ and the aircraft with the spectral feature $\mathbf{S}(r) = 0.45$ in Fig. 3 are missing targets, and the background pixel with the spectral feature $\mathbf{S}(r) = 0.86$ is a false alarm), while the fused feature subspace accurately detects targets and suppresses the background.

Algorithm 1 describes the process of the proposed TLH$^2$TD method.

### A. CTBSE Based on Transfer Learning

The RGB dataset is defined as the source domain $D_S = \{(x_{S1}, y_{S1}), \ldots, (x_{Sn}, y_{Sn})\}$, where $x_{Si} \in X_S$ is the $i$th data instance of $D_S$, $y_{Si} \in Y_S$ is the category label corresponding to $x_{Si}$, $T_S$ is the source task in the RGB target detection task, and $f_S(\cdot)$ is the source prediction function.

We define the hyperspectral data as the target domain $D_T = \{(x_{T1}, y_{T1}), \ldots, (x_{Tn}, y_{Tn})\}$, where $x_{Ti} \in X_T$ is the $i$th data instance in $D_T$, $y_{Ti} \in Y_T$ is the category label corresponding to $x_{Ti}$, $T_T$ is the source task in the HTD task, and $f_T(\cdot)$ is the source prediction function.

The purpose of transfer learning using RGB-hyperspectral data is to reduce the difference in the distribution of data features between the source and target domains. This strategy enables learning the predictive function $f_T(\cdot)$ of the target in the target domain $D_T$ using the knowledge acquired in the learning tasks $T_S$ and the prediction function $f_S(\cdot)$ of the source domain $D_S$ when only a small amount of labeled data exists in the target domain, and a sufficient amount of labeled data is available in the source domain.

The data processing steps are given as follows.

1) Preprocess the hyperspectral and high-resolution RGB data. We utilize HSIs with a few aircraft targets and a sufficient background. The HSI aircraft samples $T = \{T_i\}_{i=1}^{n}$ and the background images $I_{hb}$ are extracted from the RGB bands of HSIs. The spatial resolution of the high-resolution RGB images and HSIs is matched to enable the characterization of the target's spatial features in the HSIs, obtaining a new RGB dataset $I_R$. Since the

**Algorithm 1** Proposed TLH$^2$TD for HTD

**Input:** Original HSI $\mathbf{X}=\{\mathbf{x}_i \in \mathbb{R}^{L\times 1}\}_{i=1}^{R\times C}$, where $R$, $C$, and $L$ are the number of rows, columns, and spectral bands of the HSI.
**Output:** Final aircraft target detection result $\mathbf{D}=\{\mathrm{D}(\mathbf{r}_i)\}_{i=1}^{R\times C}$.

Initialize parameters;
**while** $trainingHTDnetwork$ **do**
  **Step1:** CTDSE-based Data Augmentation;
  Prepare a few hyperspectral target samples $\{T_1, T_2, \ldots, T_n\}$ with RGB bands and many high-resolution RGB images $\{I_1, I_2, \ldots, I_n\}$ and downsample the RGB images to match the HSI resolution;
  **for** $i \leftarrow 1$ **to** *N(N is the epoch num)* **do**
    Randomly choose 0-3 HSI target samples $T_i$ and 1 RGB image and perform an affine transformation $\hat{T}_i = affine(T_i, AffineMatrix)$;
    Apply Copy-paste, Mixup, Mosaic, etc. operations on target samples and the RGB image to expand the samples: $\hat{I}(x_{\min i} : x_{\max i}, y_{\min i} : y_{\max i}) = \alpha^* T_i + (1-\alpha)^* I(x_{\min i} : x_{\max i}, y_{\min i} : y_{\max i})$, where $\alpha$ is the Gaussian filter to smooth the edges of the pasted target;
    Obtain the current training patch image $P_i = \{I'_1, I'_2, \ldots, I'_j\}$;
  **end**
  **Step2:** HPNEM for Large-scale Complex HSIs;
  **for** $i \leftarrow 1$ **to** *N(N is the epoch num)* **do**
    Train and obtain the HTD network $N_i$ using the current patch images $I_i$;
    Calculate the loss value of all examples in the validation data;
    The results of the validation data are sorted according to the loss value. Choose positive and negative samples with the largest loss value as hard examples: $positive_i = \min(result_p)$, $negative_i = \min(result_n)$, where $positive_i : negative_i = 1 : 3$;
    Randomly copy the hard positive examples and hard negative examples to the next training patch images;
  **end**
**end**
The final training network $N$ is obtained;
**Step3:** SSWS-based Spatial-spectral Feature Fusion;
Acquire multiscale spatial features $\mathbf{V} = \{\mathbf{v}_{ti}\}_{i=1}^{3\times n}$ of the HSI using the trained network, where $\mathbf{v}_{ti} = (x, y, w, h, \theta, s)$, and each $\mathbf{v}_{ti}$ corresponds to a target;
Acquire hyperspectral features $\mathbf{S}=\{\mathrm{S}(\mathbf{r}_i)\}_{i=1}^{R\times C}$ of the HSI by calculating the Mahalanobis Distance;
Perform weighted fusion of spatial and spectral features to construct weighted spatial-spectral subspace $\mathbf{F}=\{\mathrm{F}(\mathbf{r}_i)\}_{i=1}^{R\times C}$;
Obtain the final detection result $\mathbf{D}=\{\mathrm{D}(\mathbf{r}_i)\}_{i=1}^{R\times C}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

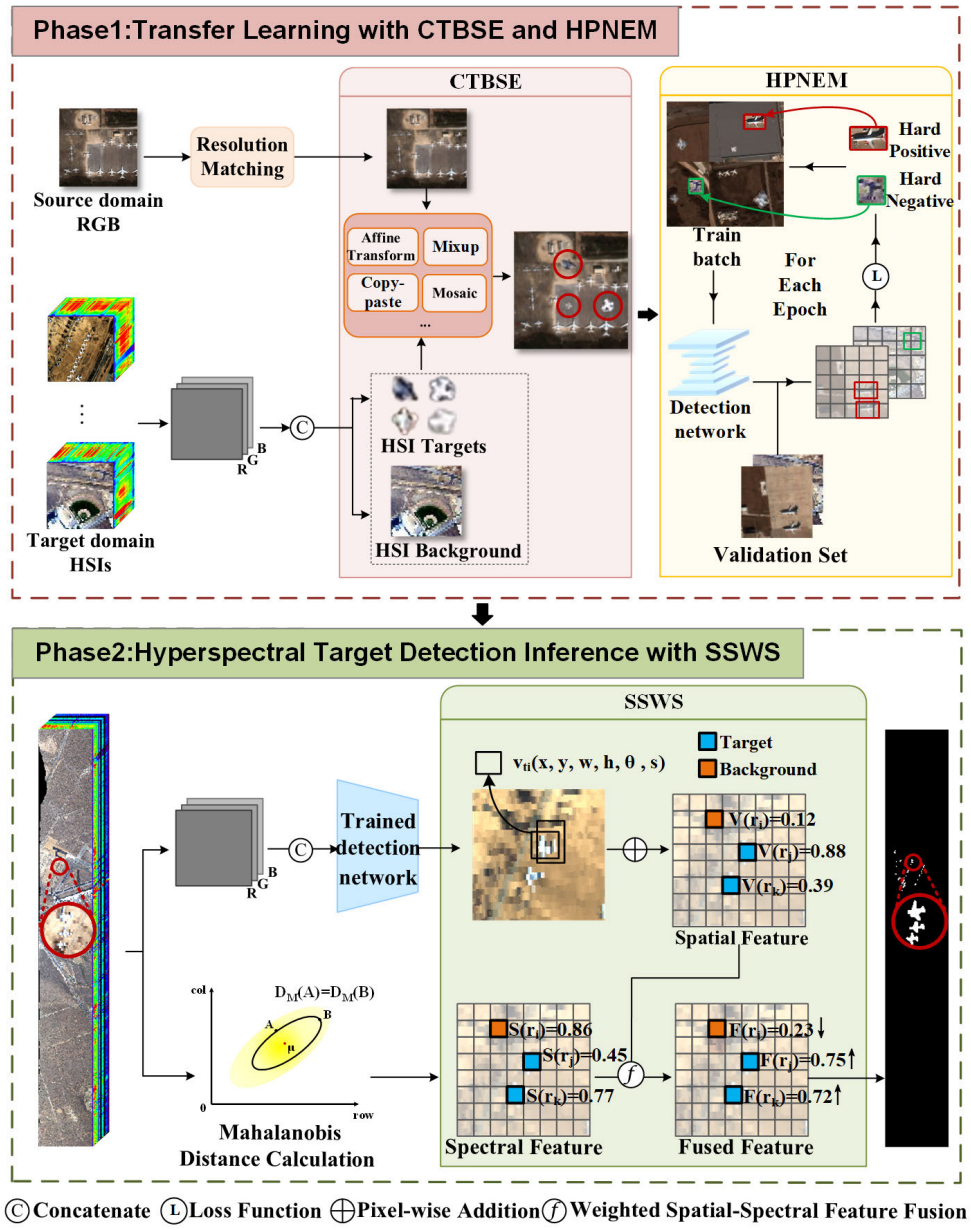



Fig. 3. Flowchart of the proposed TLH$^2$TD method for HTD.

spatial resolution of the HSI is lower than that of the RGB images, a downsampling operation is performed on the RGB dataset. The spatial features of the RGB images before and after downsampling are shown in Fig. 4.

2) Prepare training and validation data. After preprocessing the data, the RGB dataset $I_R$ is divided into $I_{R1}$ and $I_{R2}$, where $I_{R1}$ and the HSI airplane samples $T$ are sent to the target detection network as training data. $I_{R2}$ and the HSI background images $I_{hb}$ are used as validation data.
3) *CTBSE:* Before inputting the training set into each training epoch, we use an affine transformation, copy–paste [44], MixUp [45], Mosaic, and other data augmentation methods to expand the HSI target and background samples. The specific steps of the CTBSE module are given as follows.
   a) *Affine Transformation:* The target or background sample $T_i$ to be pasted is subjected to several operations, such as random scaling, rotation, and translation. The formula and transformation matrix form of the affine transformation are shown in the following equations:

$$\hat{T}_i = \text{affine}(T_i, \text{AffineMatrix}) \tag{1}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{2}$$

where $\hat{T}_i$ and $T_i$ are the samples before and after the affine transformation, respectively, AffineMatrix is the affine transformation matrix, $x$ and $y$ are the coordinates before the transformation, and $x'$ and $y'$ are the coordinates after the transformation. $m_{11}$, $m_{12}$, $m_{21}$, and $m_{22}$ are the parameters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 4. Comparison of images before and after spatial resolution matching: (a) original RGB image, (b) RGB image after resolution matching, and (c) HSI. The morphological features of the airplane in the RGB image are closer to those of the HSI after spatial resolution matching.

of the linear transformation, and $m_{31}$ and $m_{32}$ are the parameters of the translation operation.

b) *Target-Background Coupling:* The HSI target and background samples after affine transformation are pasted into the high-resolution RGB images using copy–paste, MixUp, and other methods.
The specific implementation is given as follows. For each RGB image, we randomly select 0–3 target or background samples and paste them into random positions in the image while ensuring that the intersection over union (IoU) between the minimum bounding rectangle of the pasted and existing samples does not exceed 0.3. Meanwhile, the ground-truth labels are automatically updated according to the sample positions. The center of the pasted sample $(x_{ci}, y_{ci})$ is randomly selected according to the size of the RGB image and the size of the minimum bounding rectangle $(w_i, h_i)$ of the aircraft sample, and the pasting location $(x_{\min i}{:}x_{\max i}, y_{\min i}{:}y_{\max i})$ is determined accordingly. The target-background coupling operation can be described by the following equation:

$$\hat{I}(x_{\min i} : x_{\max i}, y_{\min i} : y_{\max i}) = \alpha^* T_i + (1-\alpha)^* I(x_{\min i} : x_{\max i}, y_{\min i} : y_{\max i}) \tag{3}$$

where $T_i$ is the target or background sample being pasted, $I$ is the original image, and $\hat{I}$ is the pasted image. A Gaussian filter $\alpha$ [44] is used to perform the target-background coupling and smooth the edges of the pasted objects. The effect of target-background coupling is shown in Fig. 5.

c) *Mosaic:* The Mosaic data augmentation randomly selects four images after the above sample augmentation. It randomly crops and mosaics them to increase the richness of the samples and backgrounds and enhance model performance. The result of mosaicking is shown in Fig. 6.

The hyperspectral samples are expanded using these operations. The feature distribution of the aircraft samples in the RGB dataset is consistent with that of the HSIs.



(a) (b)

Fig. 5. Example of target-background coupling. (a) and (b) are two example images with 3 target samples pasted.



Fig. 6. Example of Mosaic data augmentation.

### B. HPNEM for Large-Scale HTD

Transfer learning significantly increases the amount of labeled aircraft samples in HSIs. However, nontargets can be misclassified as targets or vice versa in large-scale HSIs due to the sparse distribution and random location of small targets, leading to insufficient learning of positive sample representations by the model. These samples include hard positive and hard negative samples.

We utilize the idea of online hard example mining (OHEM) [46] to enable the model to learn the positive and negative sample representations near the classification threshold boundary to improve the ability to distinguish between targets and the background in large-scale complex scenes. The specific steps of the HPNEM module are given as follows.

1) In each epoch, the training batch image $I_i$ acquired from data transformation is fed into the network to obtain the training network $N_i$.
2) After the training in the current epoch, the network is forward propagated once using the validation data to obtain all candidate anchors, and the positive and negative samples are ranked based on the loss value.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

3) We select hard positive and hard negative samples with the largest loss values, i.e., samples representing the target with low confidence and those representing the background with high confidence. We maintain a ratio of positive to negative samples of 1:3 to ensure a balanced distribution of positive samples $\mathbf{P} = \{P_i\}_{i=1}^{n}$ and negative samples $\mathbf{N} = \{N_i\}_{i=1}^{m}$, where $P_i$ and $N_i$ are an instance of positive and negative samples, respectively, and $n$ and $m$ are the number of positive and negative samples.
The hard samples $\mathbf{P}$ and $\mathbf{N}$ are randomly copied into the training batch $I_i$ of the next epoch, enabling the model to learn easily misclassified targets and backgrounds. The sample copying operation can be described by the following equations:

$$\hat{I}_{ij}(x_{\min j} : x_{\max j}, y_{\min j} : y_{\max j}) = \alpha^{*} P_i + (1-\alpha)^{*} I_{ij}(x_{\min j} : x_{\max j}, y_{\min j} : y_{\max j}) \tag{4}$$

$$\hat{I}_{ij}(x_{\min j} : x_{\max j}, y_{\min j} : y_{\max j}) = \alpha^{*} N_i + (1-\alpha)^{*} I_{ij}(x_{\min j} : x_{\max j}, y_{\min j} : y_{\max j}) \tag{5}$$

where $P_i$ and $N_i$ are the hard positive and negative sample being pasted, respectively, $I_{ij}$ is the original batch image, $\hat{I}_{ij}$ is the pasted image, $(x_{\min i} : x_{\max i}, y_{\min i} : y_{\max i})$ is the pasting location of the original image, and $\alpha$ is the Gaussian filter.

4) We repeat steps 1)–3) to obtain the final trained target detection network $N$.

The hard example mining process is shown in Fig. 7. The validation data include both target and background images. After the CTBSE and network training in each epoch, the obtained network is used to perform a forward propagation on the validation data and obtain all candidate anchors, and then, the positive and negative examples are ranked based on the loss value. Positive examples represent examples that are actually targets, and negative examples represent examples that are actually background. Hard positive examples are those positive examples that have lower prediction scores and are higher losses during training, i.e., those positive examples that have been incorrectly classified as negative examples (background examples). Correspondingly, hard negative examples are those negative examples that have higher prediction scores and higher losses during training, i.e., those negative examples that have been incorrectly classified as positive examples (target examples). After ranking the positive and negative examples by loss values, respectively, we select the positive examples with the largest loss values as the hard positive examples and the negative examples with the largest loss values as the hard negative examples. In Fig. 7, positive samples with low confidence (e.g., 0.1) and negative samples with high confidence (e.g., 0.8) are selected as hard positive and negative samples, respectively. These samples are randomly replicated and fed into the next training batch until the network is fully trained. These hard positive and negative examples are difficult for the model to distinguish, so we iteratively add these hard examples into the subsequent training process to increase the training weights of the hard examples so that the model learns the target representation better.

### C. SSWS Construction for Feature Fusion

Subsequently, the spatial and spectral features of the target are extracted using the trained HTD network and the RXD algorithm based on a multivariate normal distribution. The processing steps of the SSWS module are given as follows.

1) *Spatial Feature Extraction:* The trained target detection network is used to extract the multiscale spatial features $\mathbf{V} = \{\mathbf{v}_{ti}\}_{i=1}^{3\times n}$ of HSIs, where $\mathbf{v}_{ti} = (x, y, w, h, \theta, s)$; $x$, $y$, $w$, and $h$ are the target center's horizontal and vertical coordinates, width, and height of the anchor, respectively; $\theta$ is the target category; $s$ is the confidence level; and $n$ is the number of anchors. Each $\mathbf{v}_{ti}$ corresponds to an anchor. In general, the larger the confidence level $s$, the higher the probability that it is the target category. Since there is only one category of airplane, all $\theta$ are 1, and only the confidence level $s$ is considered. To fuse the spatial and spectral features, we match the spatial features of the target to each pixel. We define the spatial feature of each pixel as the sum of the confidence levels of all detected anchors that include the pixel as follows:

$$V(\mathbf{r}_i) = V(x_i, y_i) = \sum_{j=1}^{m} s_j \tag{6}$$

where $\mathbf{r}_i$ represents a pixel, $(x_i, y_i)$ represents the position of the pixel in the images, $s_j$ is the confidence level of each anchor that includes the pixel $(x_i, y_i)$, and $m$ is the number of all anchors containing the coordinates. Therefore, the spatial feature is $\mathbf{V}' = \{V(\mathbf{r}_i)\}_{i=1}^{R\times C}$, where $\mathbf{r}_i$ represents the pixel, $R$ and $C$ are the number of rows and columns of the HSI, and $V(\mathbf{r}_i)$ is the spatial feature value of the pixel $\mathbf{r}_i$.

2) *Spectral Feature Extraction:* It is assumed that values of the target and background in the HSIs follow different normal distributions. The spectral outliers of the HSI are extracted by the RXD algorithm [8]

$$D_{\mathrm{RXD}}(\mathbf{x}) = (\mathbf{x}-\mu)^{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu) \tag{7}$$

where $x$ is a pixel vector in the image, $\mu$ is the sample mean vector, and $\Sigma$ is the sample covariance matrix of the image. The obtained spectral features are $\mathbf{S} = \{S(\mathbf{r}_i)\}_{i=1}^{R\times C}$, where $\mathbf{r}_i$ represents the pixel, $R$ and $C$ are the number of rows and columns of the HSI, respectively, and $S(\mathbf{r}_i)$ denotes the probability that the pixel is a target.

3) *Construction of Weighted Subspace of Spatial–Spectral Features:* The spectral features $\mathbf{S}$ and the spatial features $\mathbf{V}'$ are normalized, and the weighted fused subspace of the spatial–spectral features is constructed based on $S(\mathbf{r}_i)$ and $V(\mathbf{r}_i)$ at each pixel. The new weighted fused features $\mathbf{F} = \{F(\mathbf{r}_i)\}_{i=1}^{R\times C}$ are generated adaptively, where $F(\mathbf{r}_i)$ is the spatial fusion feature of each pixel. The optimal classification decision boundary is created

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

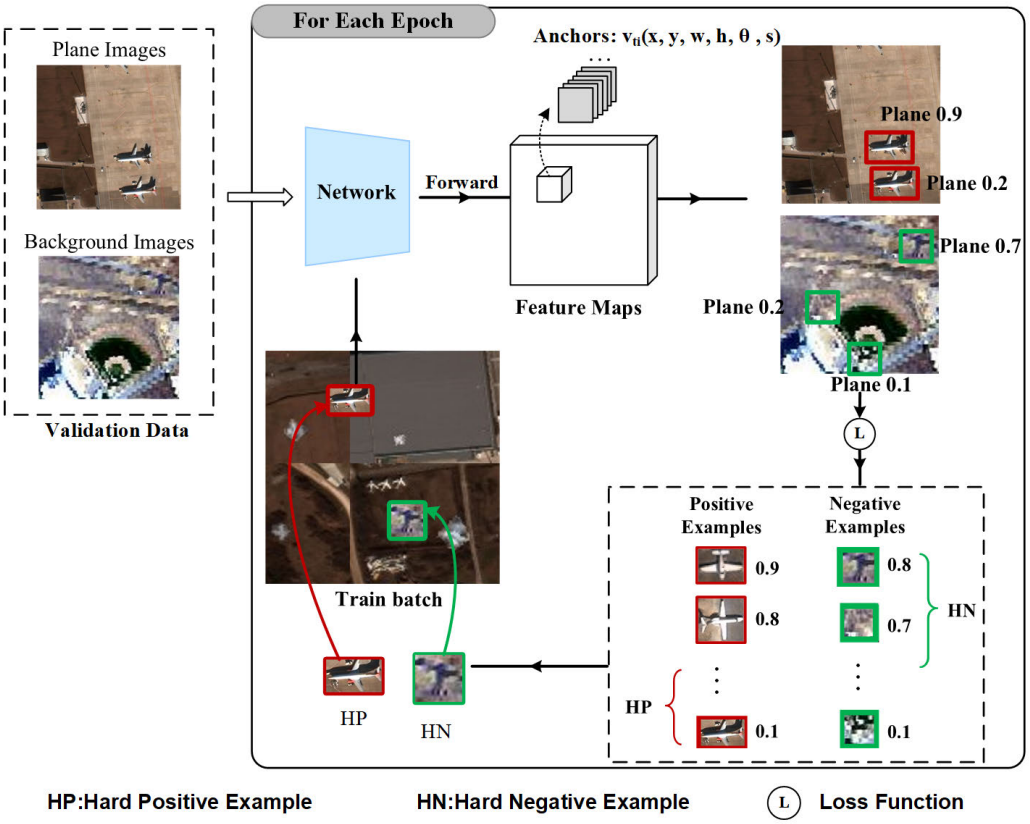



Fig. 7. Architecture of the HPNEM module.

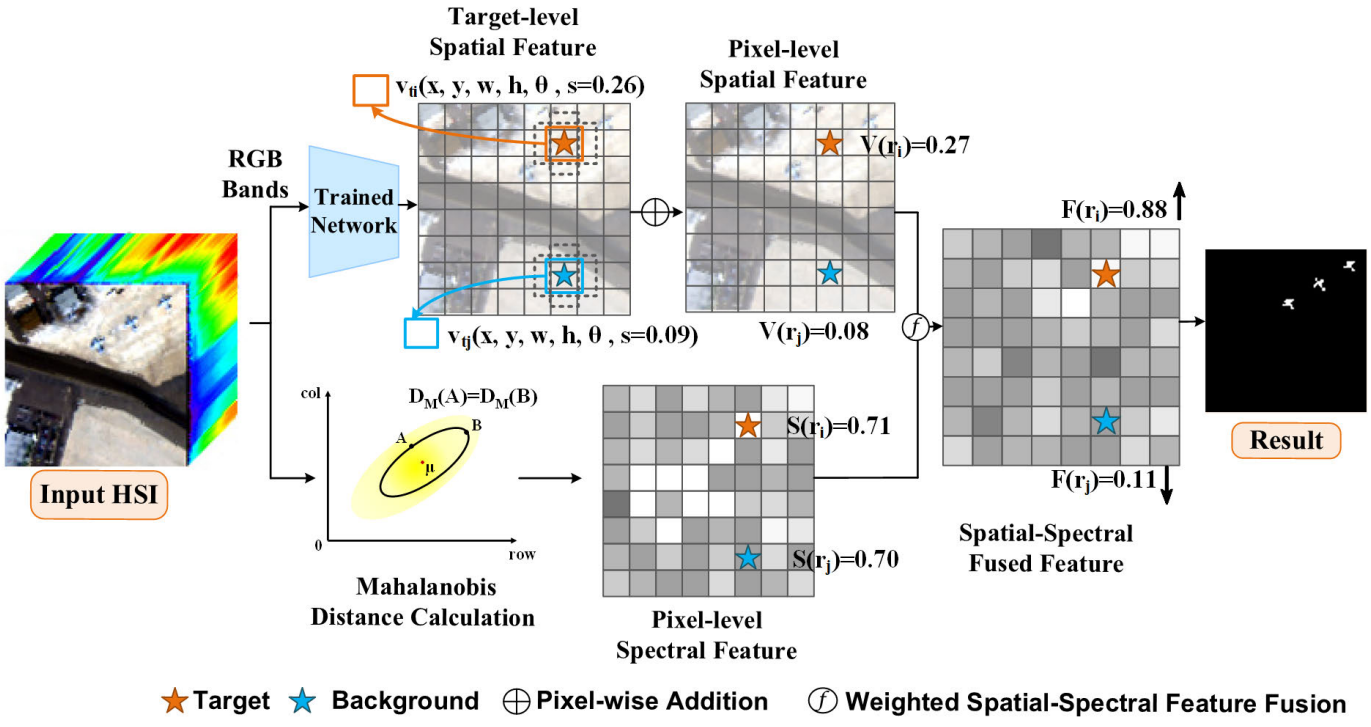



Fig. 8. Architecture of the SSWS module.

in this subspace to distinguish the target and the background to achieve high-precision HTD.

The flowchart of this process is shown in Fig. 8. This method improves the detection rate and reduces the false alarm rate. Many anchors are obtained after extracting the spatial features using the trained network. After adding and normalizing the confidence value of all anchors in each pixel, the target (i.e., confidence = 0.27)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

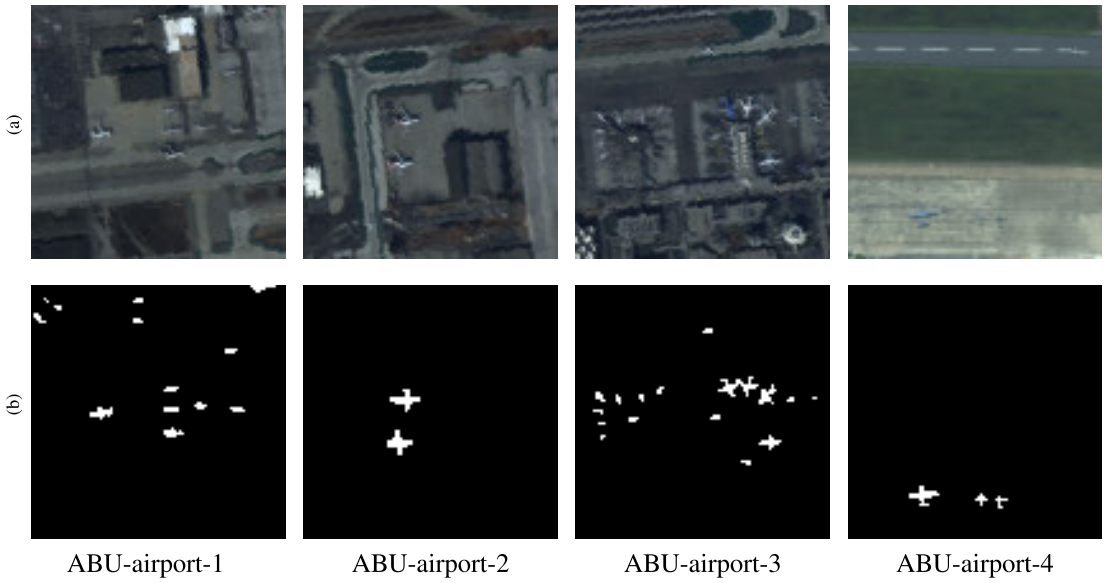



Fig. 9. ABU data: (a) true color image and (b) ground truth.

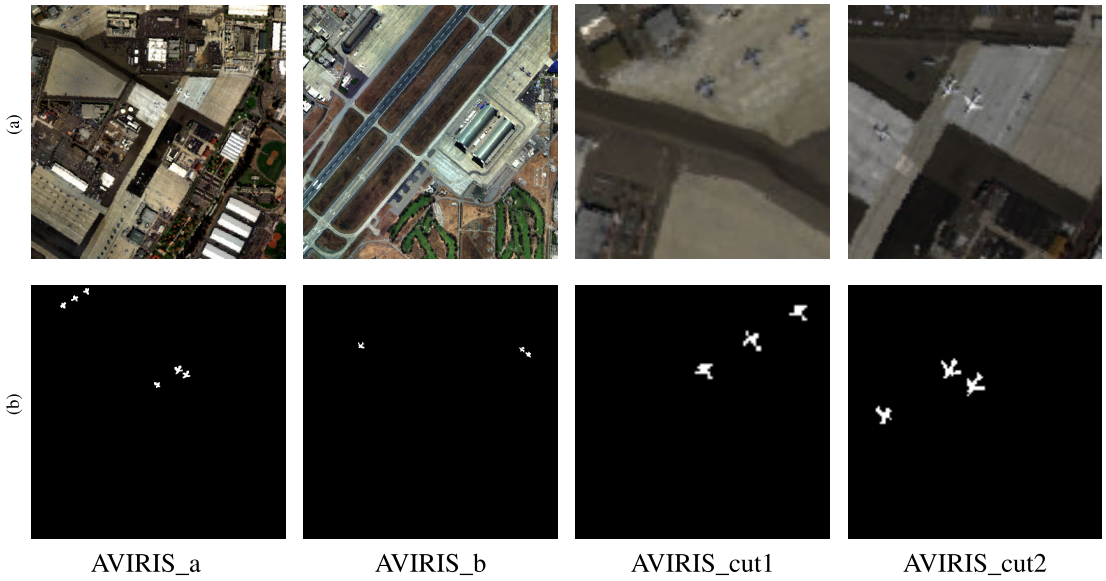


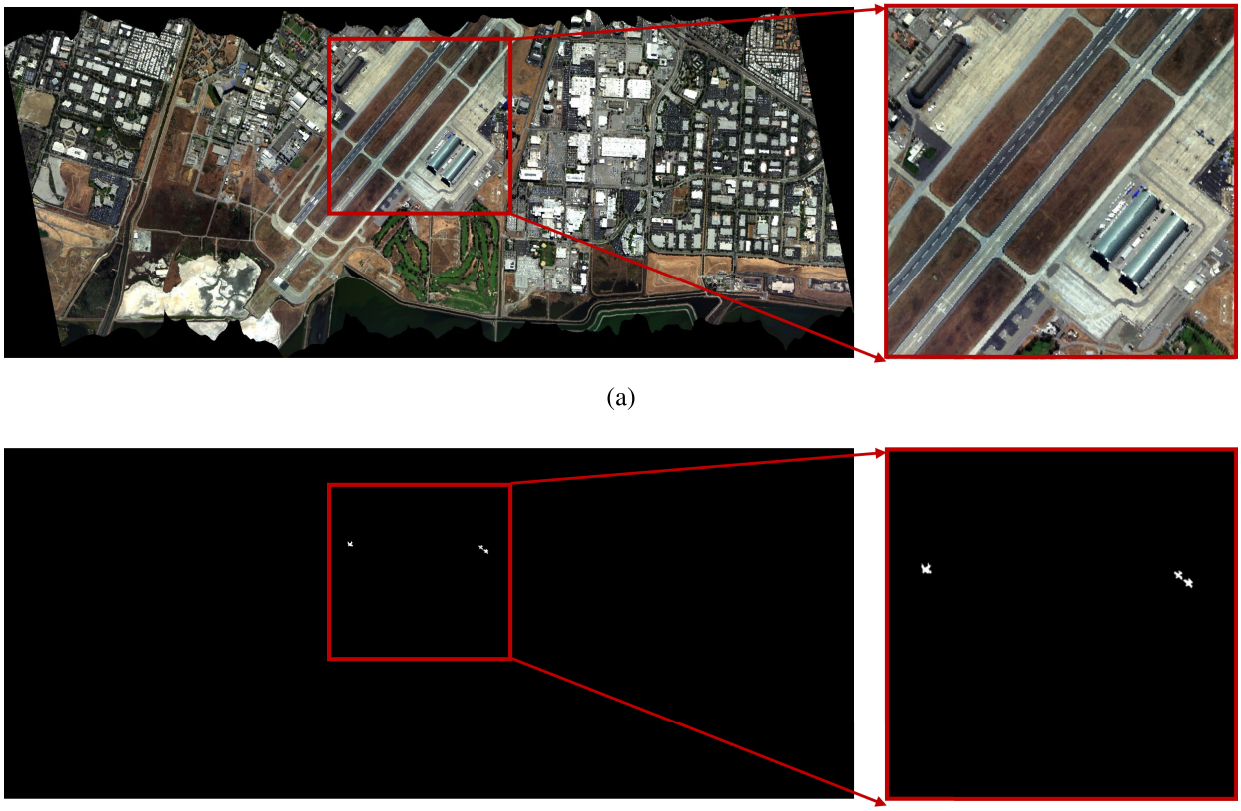

Fig. 10. Other AVIRIS data: (a) true color image and (b) ground truth.

and the background (i.e., confidence = 0.08), which both have relatively low values, are obtained. Meanwhile, the spectral eigenvalues of the target and background based on the Mahalanobis distance are 0.71 and 0.70, respectively; they are very similar. The target in Fig. 8 is easily missed using only spatial features due to its low feature value, whereas the background can be mistaken for a target using only spectral features. However, the difference between the two eigenvalues is significant after fusing the spatial and spectral features, enabling the model to distinguish the target and background and resulting in a high detection rate and low false alarm rate.

In the construction of the weighted fused space of spatial–spectral features, we define $T_s$, $Q_{s1}$, and $Q_{s3}$ as the target-background threshold, the lower quartile, and the upper quartile of the spectral feature $\mathbf{S}$, respectively; $T_v$, $Q_{v1}$, and $Q_{v3}$ are the target-background threshold, the lower quartile, and the upper quartile of the spatial feature $\mathbf{V}'$, respectively. The weighted spatial–spectral feature fusion can be expressed as follows.

1) If $S_i > T_s$ and $V_i > T_v$, the values of the spectral and spatial features are large, and the fused feature values are increased accordingly, i.e., $F_i = (w^*S_i + w^*T_i)^*8$, where the weights $w = 0.5$.
2) If $S_i < T_s$ and $V_i < T_v$, the values of the spectral and spatial features are small, and the fused feature values are reduced accordingly, i.e., $F_i = (w^*S_i + w^*T_i)/8$, where the weights $w = 0.5$.
3) If $S_i < T_s$ and $V_i > T_v$, two cases are possible.
   a) If $S_i \geq Q_{s1}$, $V_i \geq Q_{v3}$, the spectral feature values are relatively small, and the spatial feature

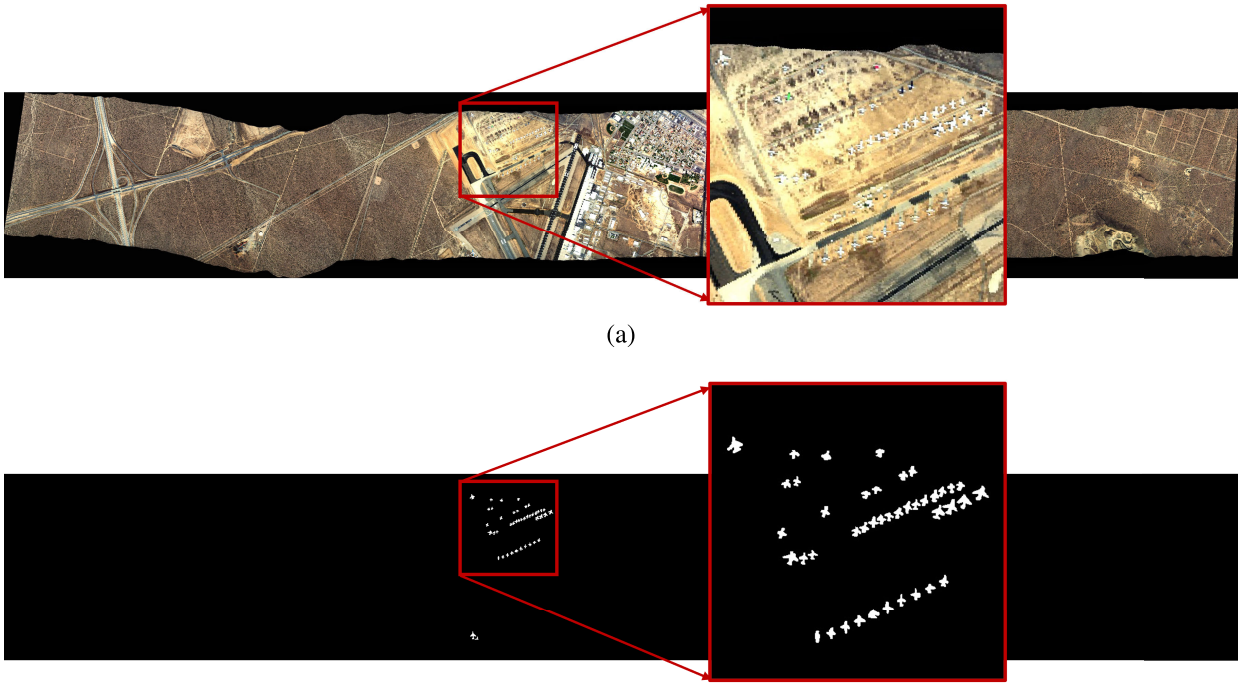
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.



(a)

(b)

Fig. 11. Large-scale hyperspectral data L-f08: (a) true color image and (b) ground truth; areas with aircraft are shown in red boxes.



(a)

(b)

Fig. 12. Large-scale hyperspectral data L-f06: (a) true color image and (b) ground truth; areas with aircraft are shown in red boxes.

values are large, we increase the fused feature values accordingly and ensure that the spatial feature weights are higher than the spectral feature weights, i.e., $F_i = (w_l * S_i + w_h * T_i) * 4$, where

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

the weights $0 < w_l < 0.5$, $0.5 < w_h < 1$, and $w_l + w_h = 1$.

b) In all other cases, the fused feature values are reduced accordingly, i.e., $F_i = (w_h * S_i + w_l * T_i)/4$, where the weights $0 < w_l < 0.5$, $0.5 < w_h < 1$, and $w_l + w_h = 1$.

4) If $S_i > T_s$ and $V_i < T_v$, two cases are possible.

a) If $S_i \geq Q_{s3}$, $V_i \geq Q_{v1}$, the spatial feature values are relatively small, and the spectral feature values are large, the fused feature values are increased accordingly to ensure that the spectral feature weights are higher than the spatial feature weights, i.e., $F_i = (w_h * S_i + w_l * T_i) * 4$, where the weights $0 < w_l < 0.5$, $0.5 < w_h < 1$, and $w_l + w_h = 1$.

b) In all other cases, the fused feature values are reduced accordingly, i.e., $F_i = (w_l * S_i + w_h * T_i) * 4$, where the weights $0 < w_l < 0.5$, $0.5 < w_h < 1$, and $w_l + w_h = 1$.

After obtaining $F_i$ corresponding to each pixel, all the values of $F_i$ in the image are normalized to [0, 1] to obtain the final spatial–spectral weighted fused feature $\mathbf{F} = \{F(\mathbf{r}_i)\}_{i=1}^{R \times C}$

$$\mathbf{F} = f(\mathbf{S}, \mathbf{V}') \tag{8}$$

where $f$ denotes the weighted fusion process. After obtaining the spatial–spectral weighted fused features, an appropriate linear decision surface is selected. The higher the value of $F(\mathbf{r}_i)$, the higher the probability that the pixel $\mathbf{r}_i$ is a target.

## IV. Experiments

### A. Datasets

The data include ten hyperspectral datasets of three different sizes and a high-resolution RGB dataset (RarePlanes). The hyperspectral datasets are used to test the proposed TLH$^2$TD, and the high-resolution RGB images are used for transfer learning of the spatial features of the small targets in the HSIs.

1) *Hyperspectral Dataset:* Following is a description of the ten hyperspectral aircraft datasets of three different sizes.

a) *ABU Dataset:* This aircraft dataset is derived from the Airport-Beach-Urban (ABU) dataset [19]. It includes four small-sized airport scenes acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. They were manually extracted from large-scale images with 205 (ABU-airport-1, ABU-airport-2, and ABU-airport-3) and 191 (ABU-airport-4) bands retained after removing the noise bands. The image size is 100 × 100 pixels, and the spatial resolutions are 7.1, 7.1, 7.1, and 3.4 m. The true color and ground-truth images are shown in Fig. 9.

b) *Other AVIRIS Datasets:* The other four hyperspectral datasets were also acquired with the AVIRIS sensor. They include two medium-sized HSIs, AVIRIS_a and AVIRIS_b, and two small-sized HSIs, AVIRIS_cut1 and AVIRIS_cut2. The latter two images were manually cropped from AVIRIS_a after removing noisy bands. AVIRIS_a has a size of 400 × 400 pixels, 224 bands, and a spatial resolution of 3.5 m. AVIRIS_b has a size of 500 × 500 pixels, 224 bands, and a spatial resolution of 3.3 m. AVIRIS_cut1 has a size of 100 × 100 pixels, 189 bands, and a spatial resolution of 3.5 m. AVIRIS_cut2 has a size of 128 × 128 pixels, 189 bands, and a spatial resolution of 3.5 m. The true color and ground-truth images are shown in Fig. 10.

c) *Large-Sized AVIRIS Datasets:* The last two hyperspectral aircraft datasets were also acquired with the AVIRIS sensor; they are uncropped large-scale HSIs f080922t01p00r03rdn_c (L-f08) and f060502t01p00r03rdn_c (L-f06). The L-f08 dataset has an image size of 2039 × 840 × 224 with a spatial resolution of 3.3 m, and L-f06 has an image size of 6122 × 922 × 224 with a spatial resolution of 2.1 m. Figs. 11 and 12 show the true color and ground-truth images of L-f08 and L-f06.

TABLE II
Details of the Experimental HSI Datasets

| HSI Dataset | Sensor | Image Size | Resolution (m) | Size Class |
|---|---|---|---|---|
| ABU-airport-1 | AVIRIS | 100*100*205 | 7.1 | Small-sized HSI |
| ABU-airport-2 | AVIRIS | 100*100*205 | 7.1 | Small-sized HSI |
| ABU-airport-3 | AVIRIS | 100*100*205 | 7.1 | Small-sized HSI |
| ABU-airport-4 | AVIRIS | 100*100*191 | 3.4 | Small-sized HSI |
| AVIRIS_cut1 | AVIRIS | 100*100*189 | 3.5 | Small-sized HSI |
| AVIRIS_cut2 | AVIRIS | 128*128*189 | 3.5 | Small-sized HSI |
| AVIRIS_a | AVIRIS | 400*400*224 | 3.5 | Medium-sized HSI |
| AVIRIS_b | AVIRIS | 500*500*224 | 3.5 | Medium-sized HSI |
| L-f08 | AVIRIS | 2039*840*224 | 3.3 | Large-sized HSI |
| L-f06 | AVIRIS | 6122*922*224 | 2.1 | Large-sized HSI |

The details of the ten datasets are listed in Table II.

2) *RarePlanes Dataset:* RarePlanes [47] is a high-resolution RGB image dataset containing real and composite satellite images from Maxar and AI.Reverie. The real part of the dataset consists of 253 Maxar WorldView-3 satellite images of 112 locations with 14 700 manually labeled aircrafts. The synthetic data were generated by AI.Reverie's simulation platform; the dataset consists of 50 000 synthetic satellite images and more than 600 000 labeled aircraft. There are 8525 images in the dataset with an image size of 512 × 512, with 5456 images in the training set, 1364 in the validation set, and 1705 in the test set.

### B. Experimental Setting

The Yolo v5x model was selected to extract spatial features after CTBSE-based data augmentation. We used 300 epochs, a batch size of 16, an initial learning rate of 0.01, and an image size of 128 × 128.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF AUC VALUES FOR TLH$^2$TD AND OTHER HTD ALGORITHMS FOR SMALL-SIZED HSIS

| Algorithms | Required a priori knowledge | AUC | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | ABU-airport-1 | ABU-airport-2 | ABU-airport-3 | ABU-airport-4 | AVIRIS_cut1 | AVIRIS_cut2 | Average |
| RXD | × | 0.8188 | 0.8377 | 0.9242 | 0.9497 | 0.8830 | 0.9152 | 0.8881 |
| Local-RX | × | **0.9400** | 0.9564 | 0.8829 | 0.9550 | 0.8066 | 0.3671 | 0.8180 |
| FrFE | × | 0.8358 | 0.9647 | 0.9358 | 0.9735 | 0.9673 | 0.9221 | 0.9332 |
| Kernel-RX | × | 0.8081 | 0.8987 | 0.8527 | 0.9513 | 0.9775 | 0.8362 | 0.8874 |
| LocalKernel-RX | × | 0.7763 | 0.7406 | 0.7005 | 0.8786 | 0.9688 | 0.8789 | 0.8240 |
| RPCA | × | 0.8970 | 0.9237 | 0.9089 | 0.9779 | 0.9747 | 0.8932 | 0.9292 |
| LRASR | × | 0.5753 | 0.7442 | 0.4353 | 0.9860 | 0.9887 | 0.7867 | 0.7527 |
| LSMAD | × | 0.8204 | 0.9218 | 0.9300 | 0.9867 | 0.9783 | 0.8899 | 0.9212 |
| SSIIFD | × | 0.8937 | 0.9358 | **0.9611** | **0.9976** | 0.9889 | 0.9432 | 0.9534 |
| PTA | × | 0.4630 | 0.6494 | 0.3288 | 0.9661 | 0.9848 | 0.7527 | 0.6908 |
| GTVLRR | Spatial | 0.8361 | 0.8203 | 0.9260 | 0.9845 | 0.9694 | 0.8275 | 0.8940 |
| KIFD | Spatial | 0.9162 | 0.9634 | 0.9530 | 0.9832 | 0.9732 | 0.9326 | 0.9536 |
| GAE | Spatial | 0.7981 | 0.9687 | 0.7919 | 0.9682 | 0.9868 | 0.9015 | 0.9025 |
| Auto-AD | Spatial | 0.6922 | 0.6741 | 0.9194 | 0.5500 | 0.2905 | 0.8336 | 0.6600 |
| RGAE | Spatial | 0.6286 | 0.7414 | 0.8749 | 0.7558 | 0.9836 | 0.9358 | 0.8200 |
| AED | Spatial | 0.9175 | 0.9690 | 0.8441 | 0.9281 | 0.8585 | **0.9512** | 0.9114 |
| ACE | Spectral | 0.7842 | 0.8449 | 0.2862 | 0.9821 | 0.9970 | 0.7712 | 0.7776 |
| CEM | Spectral | 0.8396 | 0.8993 | 0.6442 | 0.9775 | **0.9979** | 0.8385 | 0.8662 |
| TLH$^2$TD | Spatial | 0.9051 | **0.9723** | 0.9424 | 0.9882 | 0.9857 | 0.9422 | **0.9560** |

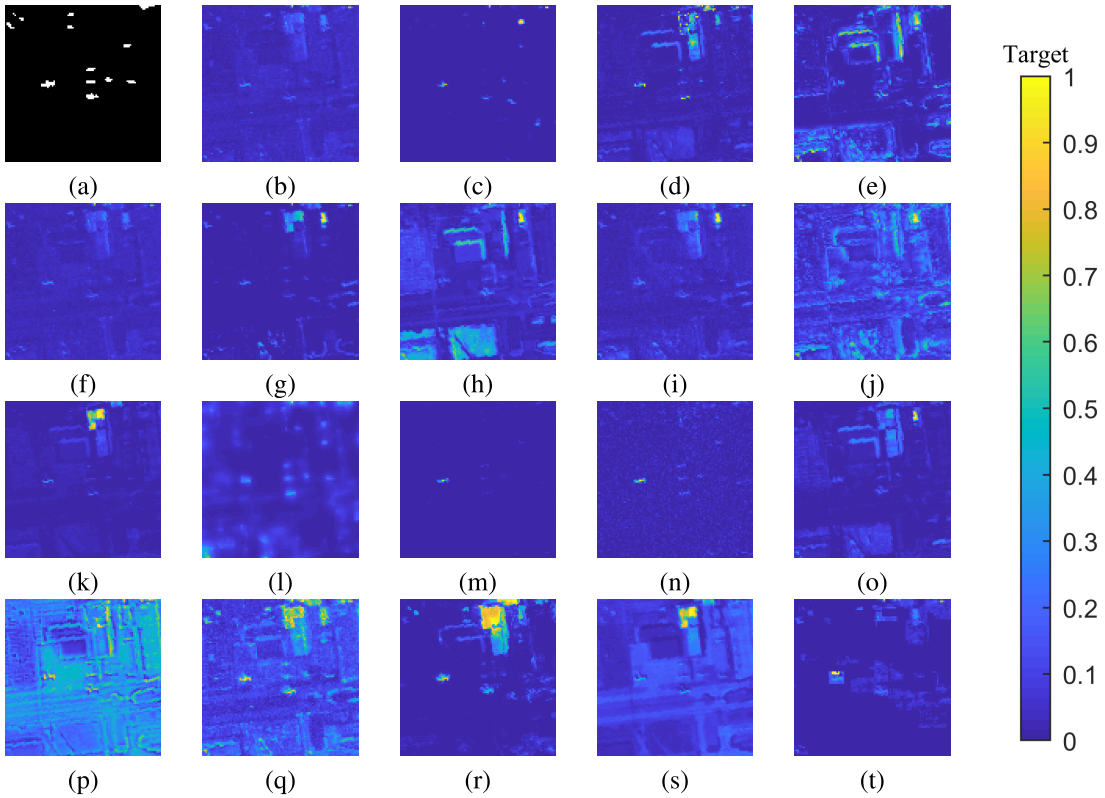



Fig. 13. Detection results of different algorithms for the ABU-airport-1 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

For ten HSIs, the detection results of the proposed method were compared with other state-of-the-art HTD algorithms. The network training and testing for extracting spatial features are carried out on a server with an Intel[1] Core[2] i7-6850K CPU, four NVIDIA TITAN Xp GPUs, and 64-GB RAM on the PyTorch framework. The construction of SSWS is implemented in the MATLAB R2018b environment.

### C. Results

The proposed TLH$^2$TD method was compared with the following HTD algorithms.

1) *RXD [8]:* The global RXD is the cornerstone algorithm in hyperspectral AED. It is a target detection algorithm

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

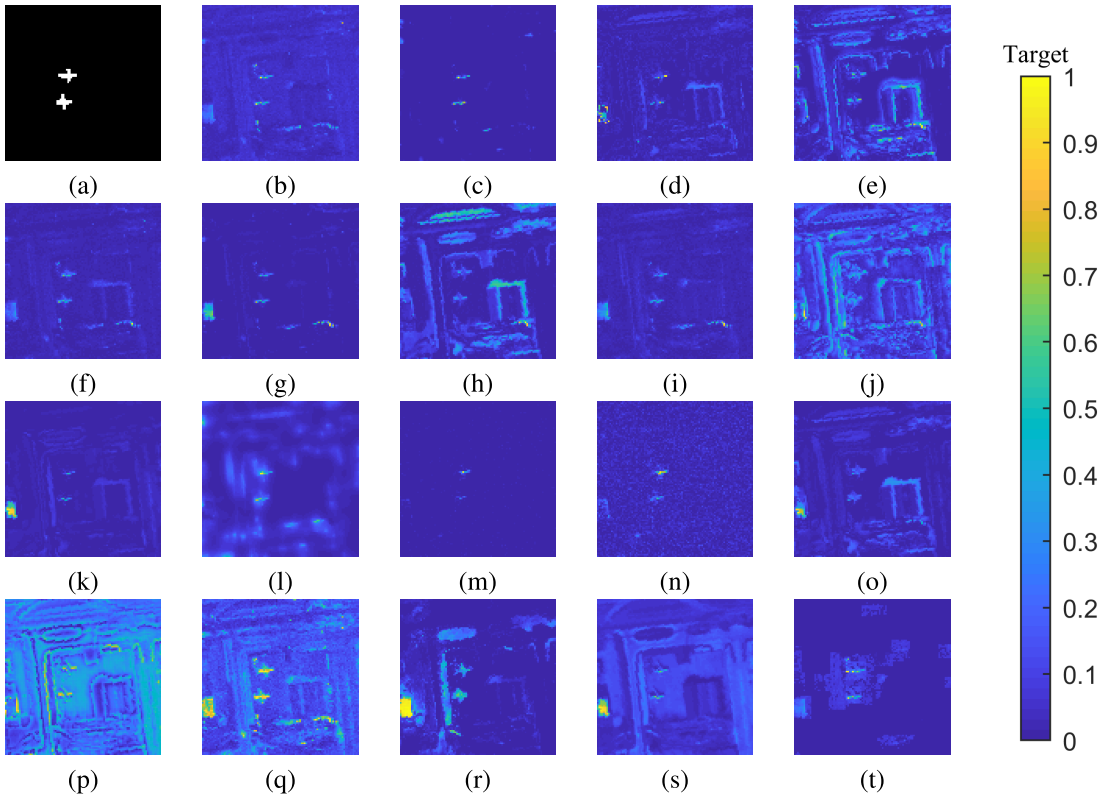



Fig. 14. Detection results of different algorithms for the ABU-airport-2 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

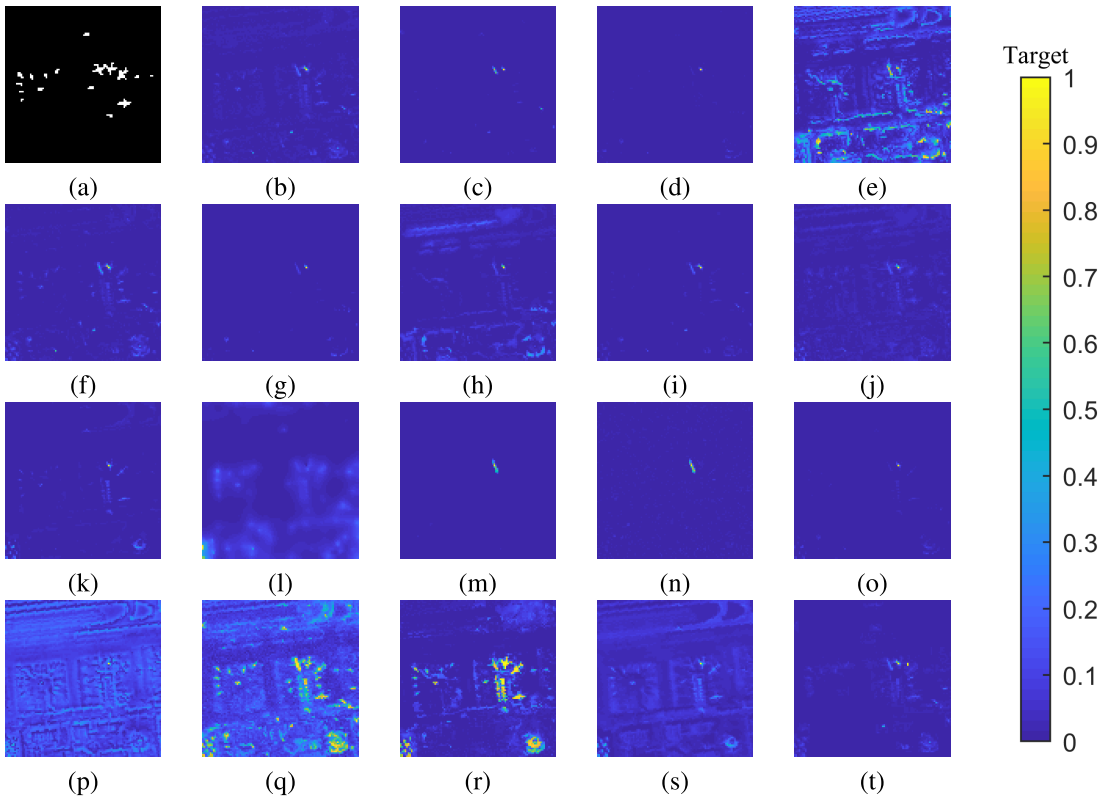



Fig. 15. Detection results of different algorithms for the ABU-airport-3 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

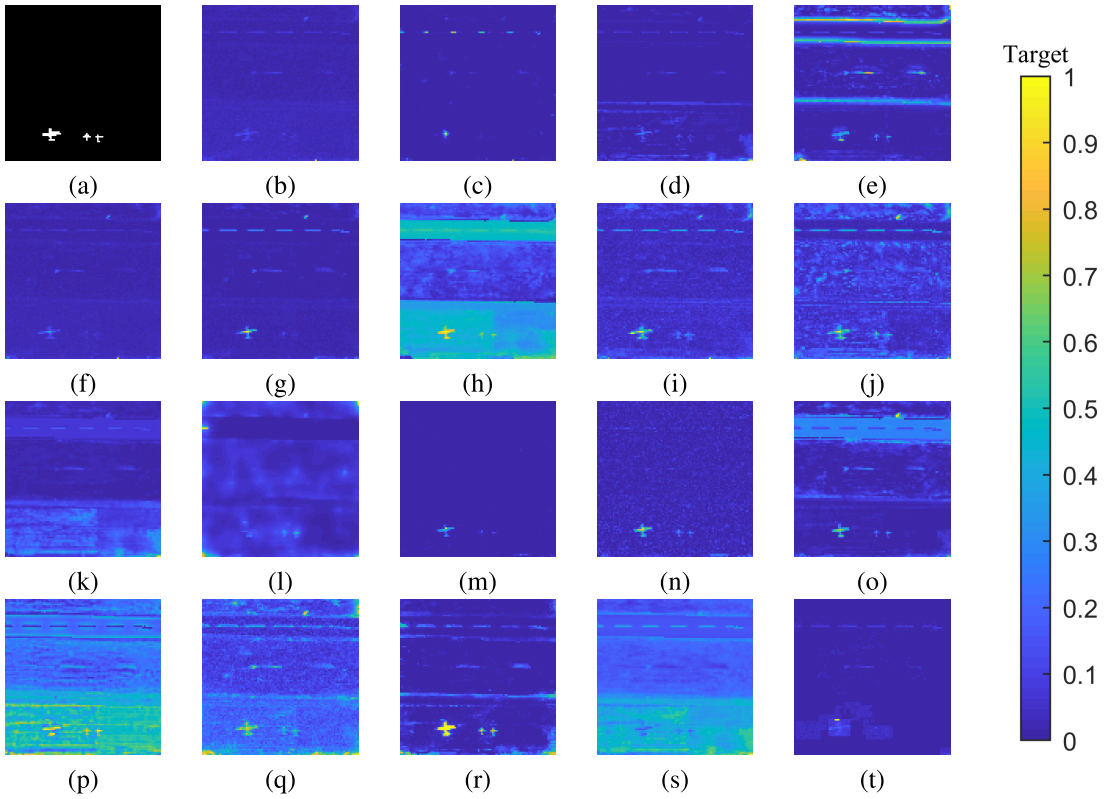



Fig. 16. Detection results of different algorithms for the ABU-airport-4 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

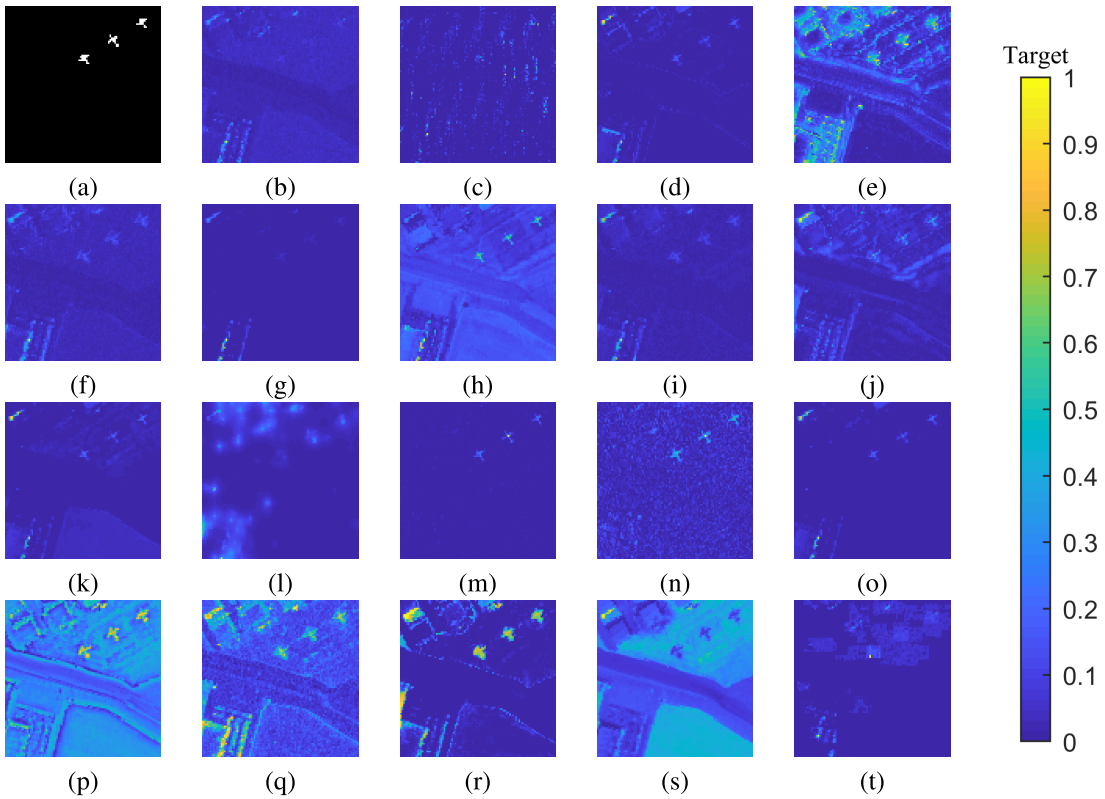



Fig. 17. Detection results of different algorithms for the AVIRIS_cut1 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

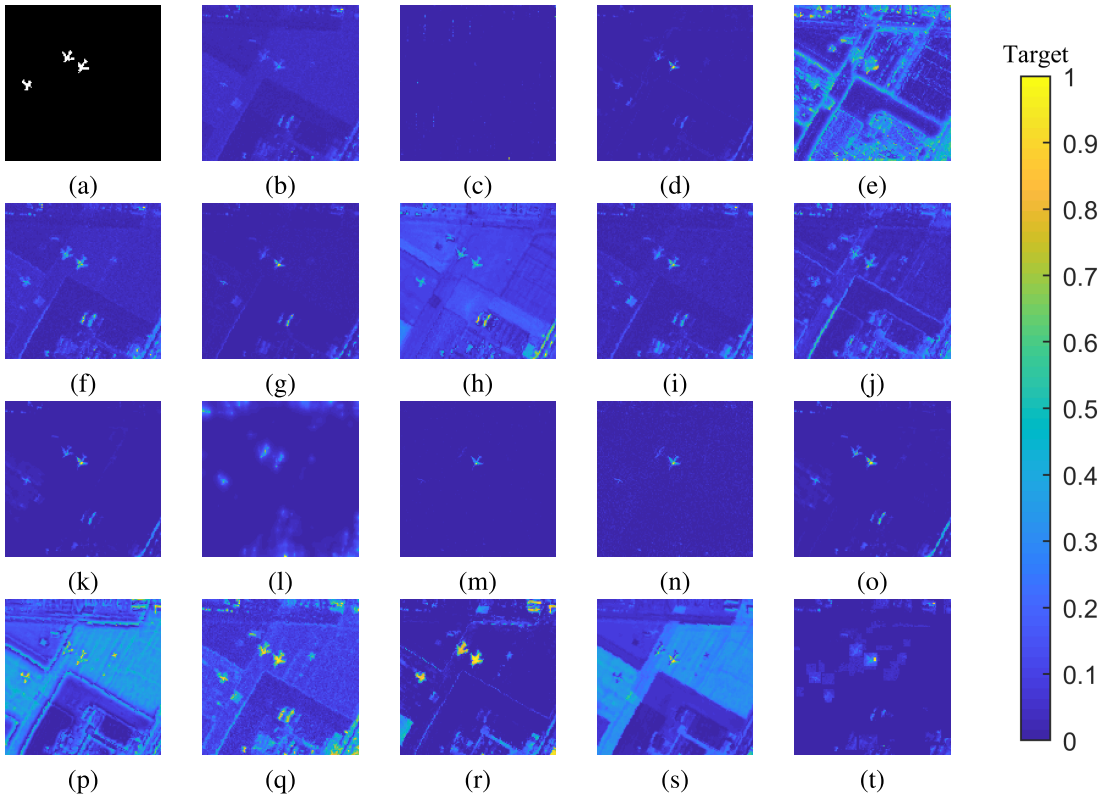



Fig. 18. Detection results of different algorithms for the AVIRIS_cut2 dataset: (a) ground truth, (b) RX, (c) local-RX, (d) KernelRX, (e) LocalKernelRX, (f) FrFE, (g) RPCA, (h) LRASR, (i) LSMAD, (j) GTVLRR, (k) RGAE, (l) AED, (m) ACE, (n) CEM, (o) GAE, (p) PTA, (q) KIFD, (r) SSIIFD, (s) AutoAD, and (t) TLH$^2$TD.

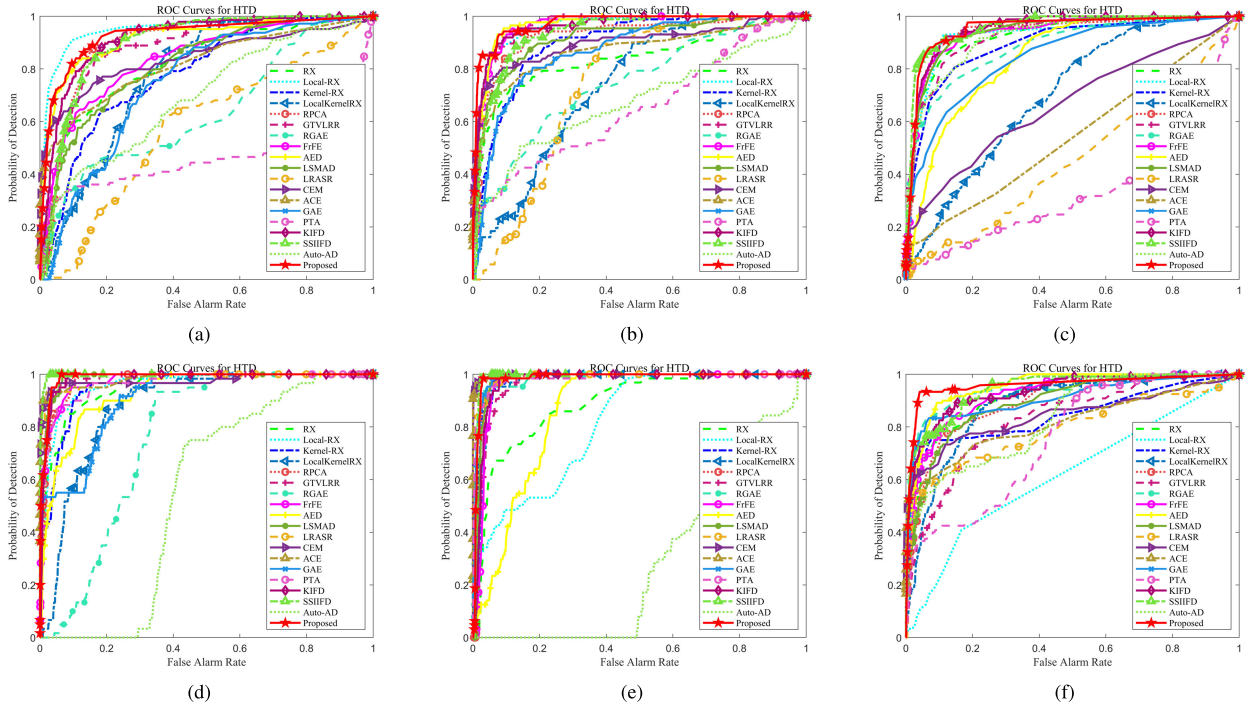



Fig. 19. Comparison of ROC for different HTD algorithms: (a) ABU-airport-1, (b) ABU-airport-2, (c) ABU-airport-3, (d) ABU-airport-4, (e) AVIRIS_cut1, and (f) AVIRIS_cut2.

that does not require prior knowledge of the target. It assumes that the values of the background have a multivariate normal distribution. It utilizes the covariance matrix to model the background and selects the anomalies using the Mahalanobis distance.

2) *Local-RX [48]:* The local RXD is a version of the RXD. The difference is whether the statistics (i.e., the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF AUC VALUES AND TIME COST FOR TLH$^2$TD AND OTHER HTD ALGORITHMS FOR MEDIUM-SIZED HSIS

| Algorithms | Required a priori knowledge | Available | AUC | | | Time Cost (second) | |
|---|---|---|---|---|---|---|---|
| | | | AVIRIS_a | AVIRIS_b | Average | AVIRIS_a | AVIRIS_b |
| RXD | × | √ | 0.7050 | 0.6777 | 0.6914 | **1.77** | 3.56 |
| Local-RX | × | √ | 0.6032 | 0.8430 | 0.7231 | 1072.36 | 3094.58 |
| FrFE | × | √ | 0.7348 | 0.8636 | 0.7992 | 256.90 | 374.86 |
| Kernel-RX | × | Array over limit | / | / | / | / | / |
| LocalKernel-RX | × | √ | 0.7842 | 0.8304 | 0.8073 | 3064.79 | 4840.72 |
| RPCA | × | Array over limit | / | / | / | / | / |
| LRASR | × | Out of memory | / | / | / | / | / |
| LSMAD | × | Out of memory | / | / | / | / | / |
| SSIIFD | × | √ | 0.7233 | 0.8814 | 0.8024 | 314.16 | 540.95 |
| PTA | × | √ | 0.6623 | 0.4085 | 0.5354 | 646.81 | 1182.61 |
| GTVLRR | Spatial | √ | 0.7549 | 0.9311 | 0.8430 | 1651.87 | 3550.87 |
| KIFD | Spatial | Out of memory | / | / | / | / | / |
| GAE | Spatial | √ | 0.3267 | 0.6368 | 0.4818 | 1091.92 | 1603.97 |
| Auto-AD | Spatial | √ | 0.6025 | 0.4343 | 0.5184 | 31.91 | 49.07 |
| RGAE | Spatial | √ | 0.5269 | 0.7587 | 0.6428 | 2279.53 | 3781.44 |
| AED | Spatial | √ | 0.7910 | 0.7922 | 0.7916 | 3.60 | 4.73 |
| ACE | Spectral | √ | 0.6120 | 0.6498 | 0.6309 | 3.70 | 4.66 |
| CEM | Spectral | √ | 0.6620 | 0.6580 | 0.6600 | 2.02 | **2.30** |
| TLH$^2$TD | Spatial | √ | **0.9776** | **0.9355** | **0.9566** | 1.90 | 3.66 |

mean and covariance matrices) are obtained by sampling all pixels in the image or pixels in a localized region surrounding the target. This algorithm does not require prior knowledge of the target.

3) *FrFE [14]:* The FrFE algorithm uses the fractional domain Fourier transform (FrFT) to enhance the signal while suppressing the noise and obtains the intermediate domain features between the original reflectance spectra and the Fourier transform through a spatial frequency representation that is complementary. It does not require prior knowledge of the target.
4) *Kernel-RX [11]:* The kernel-RX algorithm is a global RXD-derived algorithm that utilizes a kernel function to project the data into a higher dimensional space, resulting in a better separation of the background and the target. It does not require prior knowledge of the target.
5) *LocalKernel-RX [11]:* LocalKernel-RX is based on kernel-RX and utilizes two sliding windows to reduce the dimension of the kernel covariance matrix. It uses the image pixels between the two sliding windows as the local background data. It does not require prior knowledge of the target.
6) *RPCA [49]:* Robust principal component analysis (RPCA) decomposes an HSI matrix into a low-rank background matrix and a sparse target matrix for target detection. It does not require prior knowledge of the target.
7) *LRASR [50]:* Low-rank and sparse representation (LRASR) is a hyperspectral AED that models the background with low-rank representations. It improves the local representation of image pixels by adding a regularization term that induces sparsity. It does not require prior knowledge of the target.
8) *LSMAD [51]:* The low-rank and sparse matrix decomposition (LSMAD) algorithm distinguishes the background from the target and utilizes low-level a priori knowledge of the background. The target is detected using the Mahalanobis distance. It does not require prior knowledge of the target.
9) *SSIIFD [43]:* The SSIIFD algorithm is a spectral–spatial anomaly hyperspectral detector based on an improved isolation forest (iForest) to make full use of global and local information, as well as spectral and spatial information. First, it used the Gabor filter and the relative mass iForest (ReMass-iForest) detector to obtain the spatial anomaly score. Then, the spectral anomaly score was obtained by the ERS algorithm and the proposed improved iForest (IIF) detector. Finally, it fused the spatial and spectral anomaly scores by combining them linearly to predict anomaly pixels. It does not require prior knowledge of the target.
10) *PTA [52]:* The PTA algorithm is a prior-based tensor approximation for hyperspectral AED, in which HSI is decomposed into a background tensor and an anomaly tensor. In the background tensor, a low-rank prior is incorporated into spectral dimension by truncated nuclear norm regularization, and a piecewise-smooth prior on spatial dimension can be embedded by linear total variation-norm regularization. For anomaly tensor, it is unfolded along spectral dimension coupled with spatial group sparse prior that can be represented by the $l_{2,1}$-norm regularization. All the priors were integrated into a unified convex optimization framework, and the detection map was derived when the iterations of convex

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

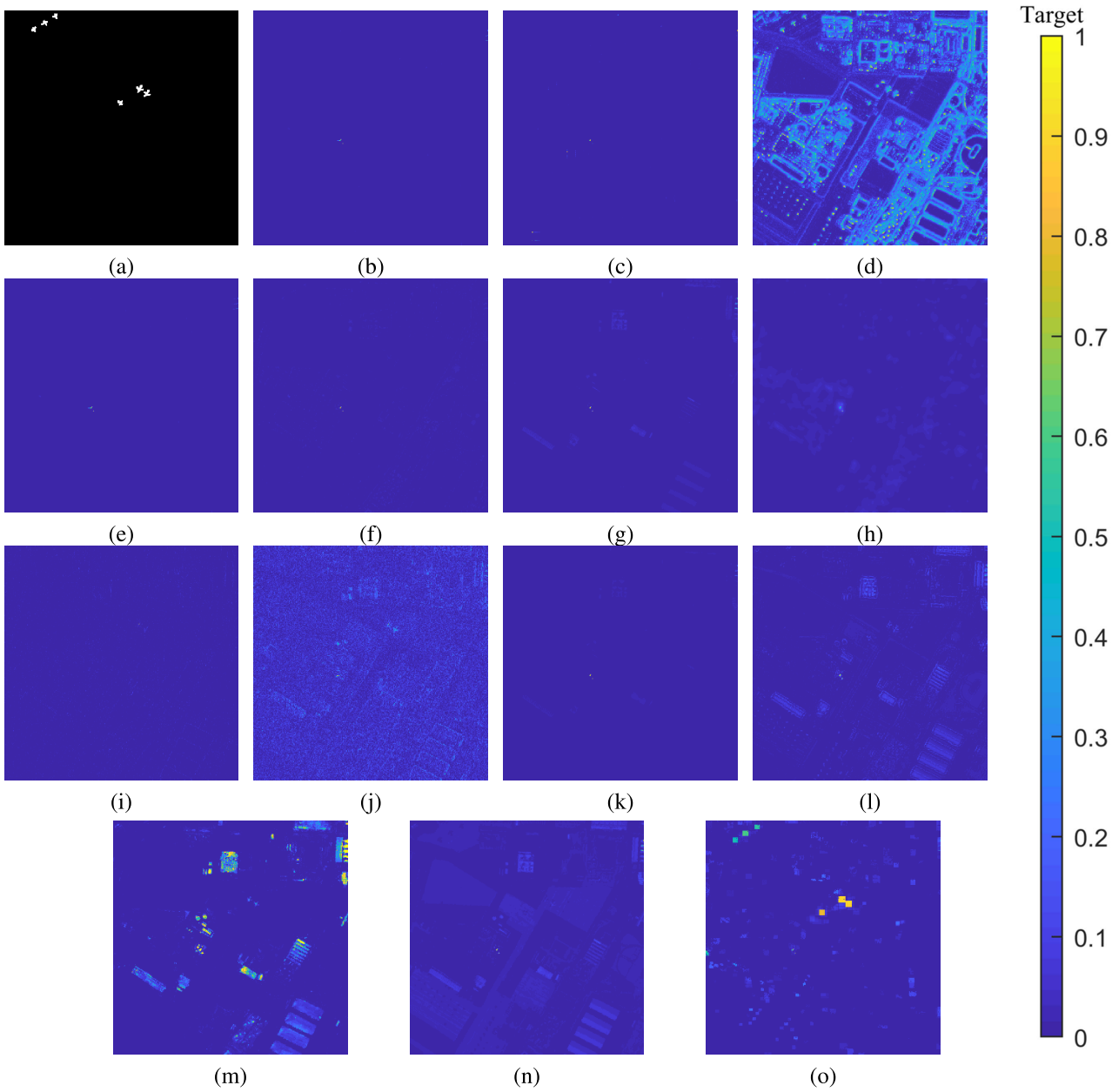



Fig. 20. Detection results of different algorithms for the AVIRIS_a dataset: (a) ground truth, (b) RXD, (c) local-RX, (d) LocalKernelRX, (e) FrFE, (f) GTVLRR, (g) RGAE, (h) AED, (i) ACE, (j) CEM, (k) GAE, (l) PTA, (m) SSIIFD, (n) AutoAD, and (o) TLH$^2$TD.

optimization converged. It does not require prior knowledge of the target.

11) *GTVLRR [53]:* The graph and total variation regularized low-rank representation (GTVLRR) model preserves the local geometric structure and spatial relationships in HSIs to enhance AED. It requires the geometric properties of the target as prior spatial knowledge of the target.
12) *KIFD [54]:* The KIFD algorithm is based on kernel iForest. The method assumes that anomalies rather than background can be more susceptible to isolation in the kernel space. Based on this idea, the proposed method first mapped the hyperspectral data into the kernel space and chose the first $K$ principal components. Then, the isolation samples in the image are detected with the iForest constructed using randomly selected samples in the principal components. Finally, the initial AED map is iteratively refined with locally constructed iForest in connected regions with large areas. The method uses the area of targets as prior knowledge and requires a prior spatial knowledge of targets.
13) *GAE [55]:* The GAE algorithm is based on a guided autoencoder to reduce the feature representation for the anomaly targets. It used a multilayer autoencoder network with skip connections to extract the abundant latent features from HSIs and then used a guided image calculated by the local spatial features of the HSIs to suppress the anomaly targets in the obtained reconstructed HSI and better represent background features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

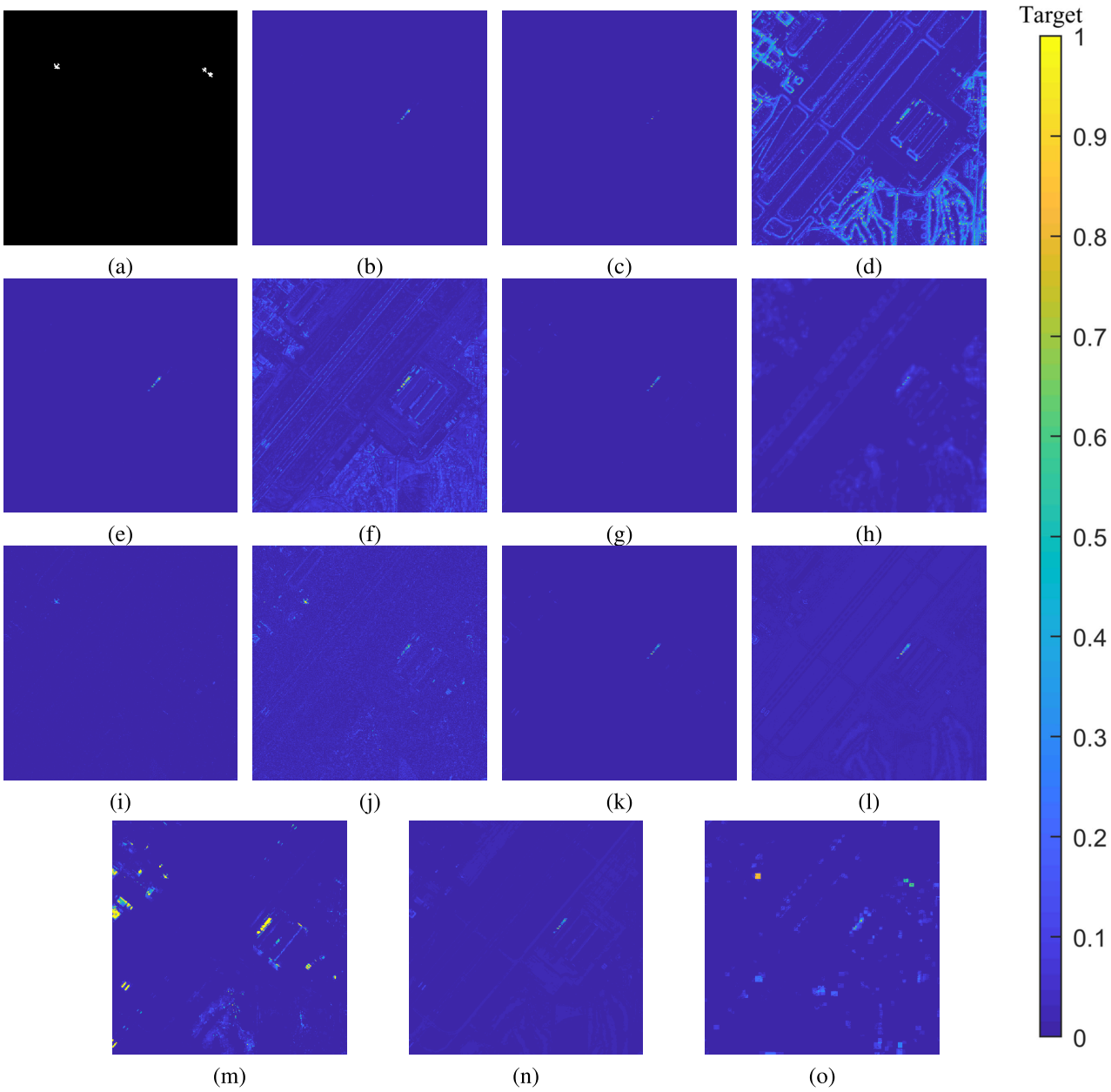



Fig. 21. Detection results of different algorithms for the AVIRIS_b dataset: (a) ground truth, (b) RXD, (c) local-RX, (d) LocalKernelRX, (e) FrFE, (f) GTVLRR, (g) RGAE, (h) AED, (i) ACE, (j) CEM, (k) GAE, (l) PTA, (m) SSIIFD, (n) AutoAD, and (o) TLH$^2$TD.

It requires the local spatial features of the target as prior spatial knowledge of the target.

14) *Auto-AD [56]:* The Auto-AD algorithm is an autonomous hyperspectral AED network (Auto-AD), in which the background is reconstructed well and the anomalies appear as reconstruction errors. The method reconstructed the background through a fully convolutional autoencoder with skip connections. Then, to further suppress the anomaly reconstruction, it designed an adaptive-weighted loss function to reduce the weights of potential anomalous pixels with large reconstruction errors during training. It assumes that the targets are relatively small compared to the background and requires the area of targets as prior spatial knowledge of the target.

15) *RGAE [57]:* The robust graph autoencoder (RGAE) is a robust anomaly detector based on an autoencoder framework that is robust to noise and anomalies during training. The autoencoder contains a graph regularization term (SuperGraph) based on superpixel segmentation to maintain the geometric structure and local spatial consistency of the HSI and reduce the per-pixel search space and execution time. It requires the geometric properties of the target as prior spatial knowledge of the target.

16) *AED [19]:* The AED algorithm performs initial pixel detection using spectral features and area attributes and then improves the initial detection results with edge-preserving filtering, considering the spatial correlation between neighboring pixels. It requires the area

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

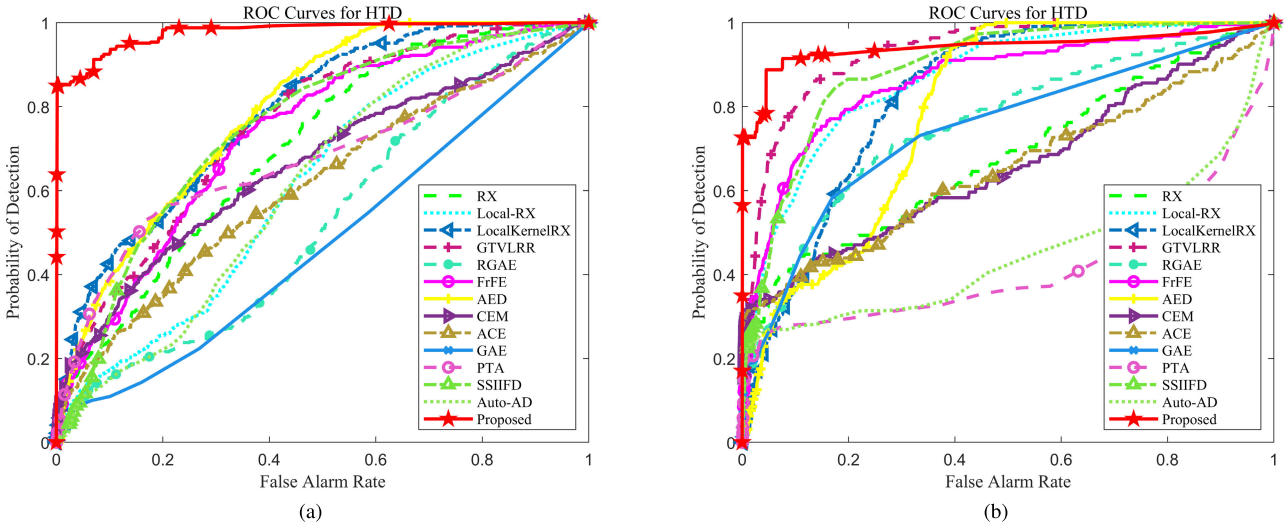



Fig. 22. Comparison of ROC for different HTD algorithms: (a) AVIRIS_a and (b) AVIRIS_b.

TABLE V
COMPARISON OF AUC VALUES AND TIME COST FOR TLH$^2$TD AND OTHER HTD ALGORITHMS FOR LARGE-SIZED HSIS

| Algorithms | Required a priori knowledge | Available | AUC | | | Time Cost (second) | |
|---|---|---|---|---|---|---|---|
| | | | L-f08 | L-f06 | Average | L-f08 | L-f06 |
| RXD | × | √ | 0.6575 | 0.9393 | 0.7984 | 33.32 | 30.11 |
| Local-RX | × | Array over limit | / | / | / | / | / |
| FrFE | × | √ | 0.7905 | 0.9262 | 0.8584 | 2576.49 | 4118.09 |
| Kernel-RX | × | Array over limit | / | / | / | / | / |
| LocalKernel-RX | × | Timeout(>9h) | / | / | / | / | / |
| RPCA | × | Array over limit | / | / | / | / | / |
| LRASR | × | Out of memory | / | / | / | / | / |
| LSMAD | × | Out of memory | / | / | / | / | / |
| SSIIFD | × | Out of memory | / | / | / | / | / |
| PTA | × | Out of memory | / | / | / | / | / |
| GTVLRR | Spatial | Timeout(>9h) | / | / | / | / | / |
| KIFD | Spatial | Out of memory | / | / | / | / | / |
| GAE | Spatial | Out of memory | / | / | / | / | / |
| Auto-AD | Spatial | Out of memory | / | / | / | / | / |
| RGAE | Spatial | Out of memory | / | / | / | / | / |
| AED | Spatial | √ | 0.6834 | 0.6532 | 0.6683 | 37.76 | 75.02 |
| ACE | Spectral | √ | 0.7839 | 0.7992 | 0.7916 | 36.48 | 52.08 |
| CEM | Spectral | √ | 0.7700 | 0.5987 | 0.6844 | **16.20** | **23.63** |
| TLH$^2$TD | Spatial | √ | **0.8992** | **0.9537** | **0.9265** | 33.88 | 30.77 |

attribute of the target as prior spatial knowledge of the target.

17) *ACE [16]:* The ACE algorithm assumes that the target and background follow different probabilistic models. It uses the generalized likelihood ratio discriminant to acquire detectors. It requires a prior spectral knowledge of the target.
18) *CEM [17]:* The CEM algorithm is derived from the linearly constrained minimum variance beamformer used in digital signal processing. It extracts signals in a particular direction while suppressing interference from signals in other directions. It requires a prior spectral knowledge of the target.

*1) Results of TLH$^2$TD and Other HTD Algorithms for Small-Sized HSIs:* Table III lists the area under the curve (AUC) values for TLH$^2$TD and other HTD algorithms.

Figs. 13–19 show the target detection results and receiver operating characteristics (ROC) curve. In these results figures, the subplot "(a) Ground truth" is a binarized image while other subplots are probability plots of the detection results. To be more specific, the pixels of the subplot (a) include only the categories of either target (the value is 1 and the color is white) or background (the value is 0 and the color is black), whereas the pixels of other subplots include probability values that vary discretely from 0 to 1 with the colors ranging from dark blue to light yellow. For these subplots, darker colors represent a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

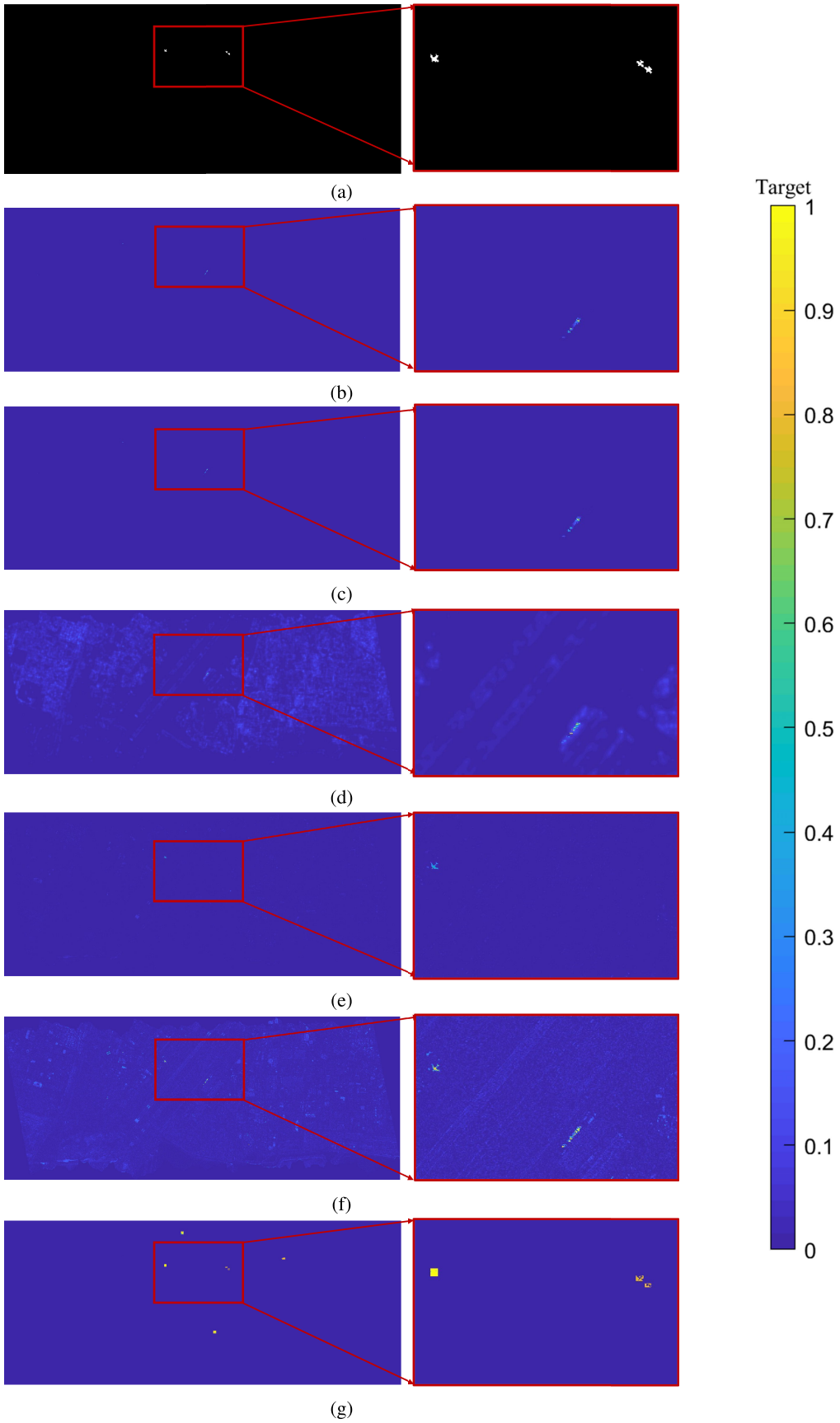



Fig. 23. Detection results of different algorithms for the L-f08 dataset: (a) ground truth, (b) RXD, (c) FrFE, (d) AED, (e) ACE, (f) CEM, and (g) TLH$^2$TD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

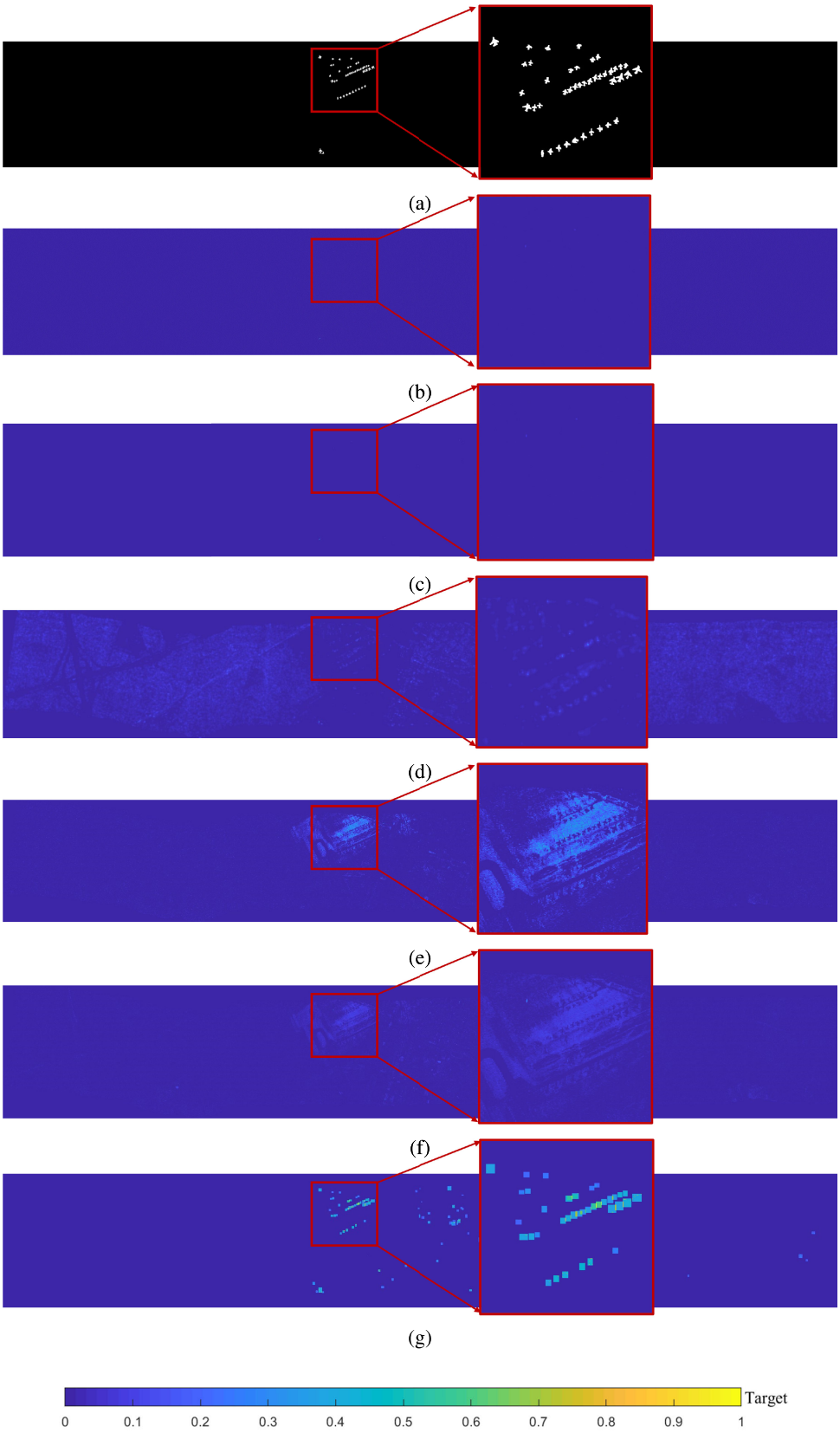



Fig. 24. Detection results of different algorithms for the L-f06 dataset: (a) ground truth, (b) RXD, (c) FrFE, (d) AED, (e) ACE, (f) CEM, and (g) TLH$^2$TD.

higher probability that the pixel is the background, and lighter colors represent a higher probability that the pixel is the target.

The results indicate that the average AUC values obtained by the proposed TLH$^2$TD are higher ($>$0.9) than those of the other methods. Although the local-RX algorithm has a higher AUC value for the ABU-airport-1 data, the AUC value on AVIRIS_cut2 is very low (0.3671). The AED algorithm has a higher AUC value on AVIRIS_cut2, but the AUC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

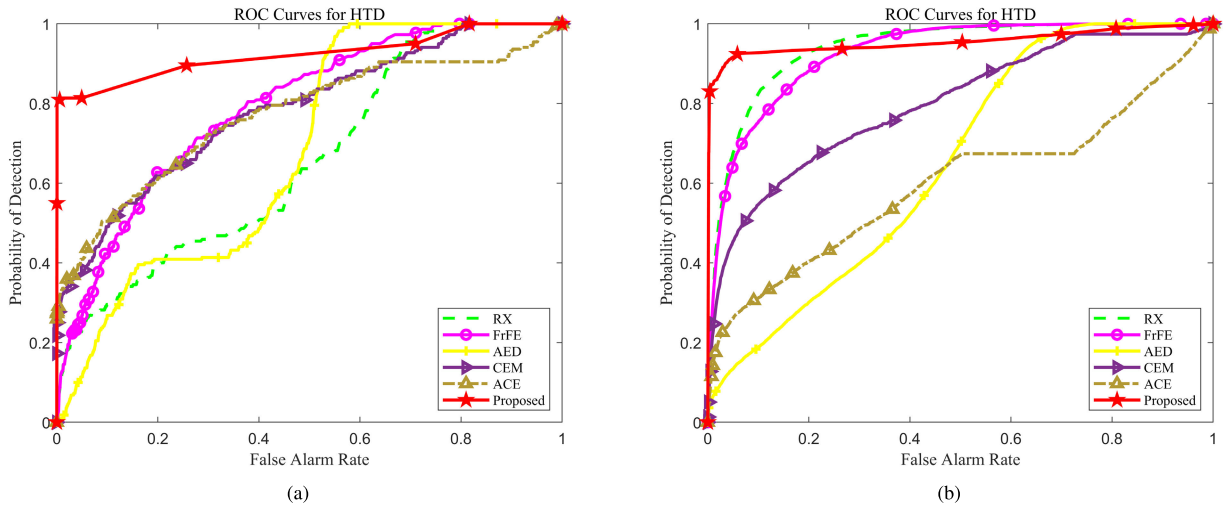



Fig. 25. Comparison of ROC for different HTD algorithms: (a) L-f08 and (b) L-f06.

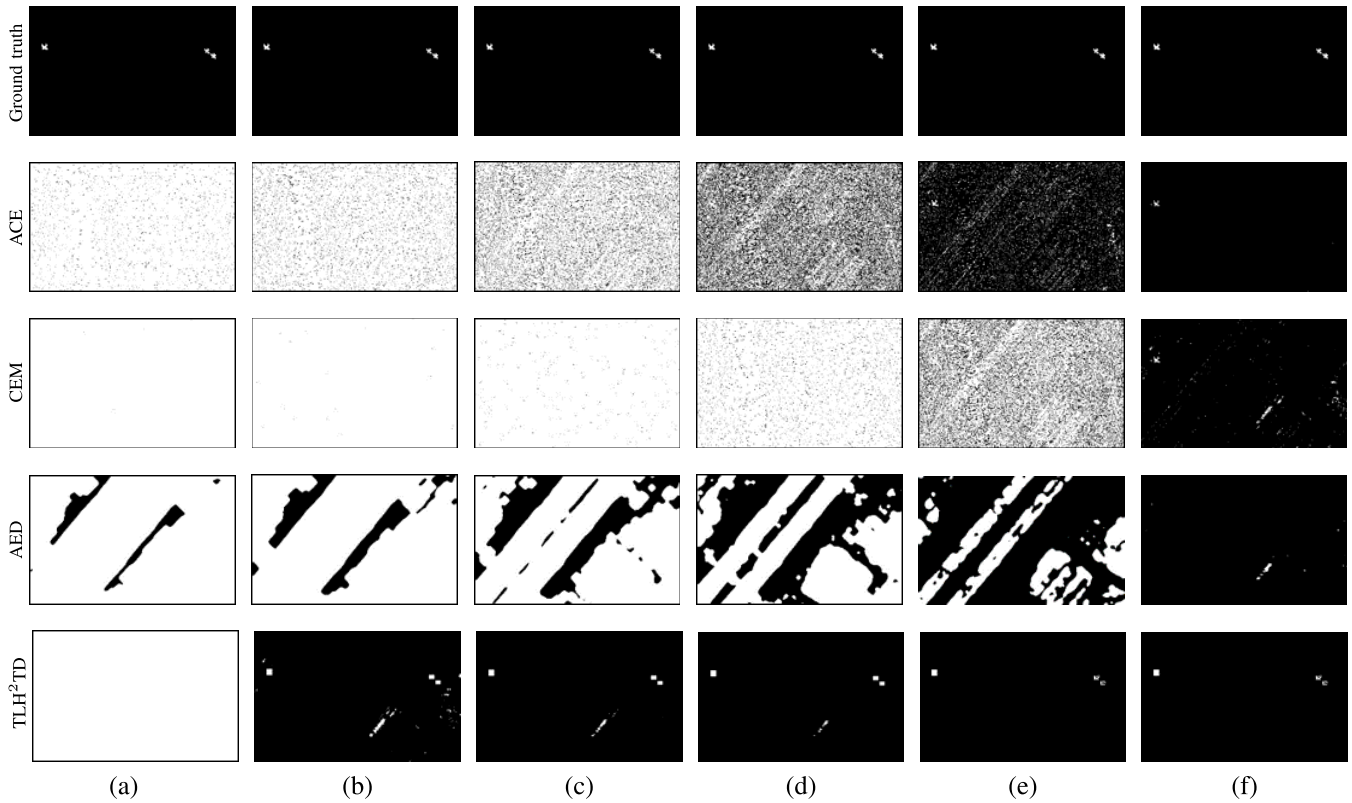



Fig. 26. Comparison of binarization results of different HTD algorithms for the L-f08 dataset: (a) thre = 1e-6, (b) thre = 1e-5, (c) thre = 1e-4, (d) thre = 1e-3, (e) thre = 1e-2, and (f) thre = 0.1. Note that thre denotes the threshold for target and background segmentation.

values for the other five datasets are lower than that of the TLH$^2$TD. Especially, the AUC values on ABU-airport-3 and AVIRIS_cut1 are only around 0.85. The average AUC values of the PTA and Auto-AD algorithms are lower than 0.7, indicating that they are not robust. Though the GAE algorithm has quite high average AUC values of more than 0.9, the AUC values of ABU-airport-1 and ABU-airport-3 are lower than 0.8. The CEM algorithm has a slightly higher AUC value than TLH$^2$TD on AVIRIS_cut1, but the AUC value on the ABU-airport-3 dataset is only 0.6442, indicating that the algorithm has failed. The SSIIFD and KIFD algorithms all have an average AUC value of more than 0.9, but the AUC of SSIIFD on ABU-airport-1 is less than 0.9, and only two out of six AUC values of KIFD were higher than TLH$^2$TD. The comprehensive comparison indicates that the proposed method is more robust than the other algorithms and applies to a larger variety of hyperspectral scenes.

*2) Results of TLH$^2$TD and Other HTD Algorithms for Medium-Sized HSIs:* Comparative experiments are performed on two medium-sized HSIs, AVIRIS_a (400 × 400 × 224) and AVIRIS_b (500 × 500 × 224) with TLH$^2$TD, and other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

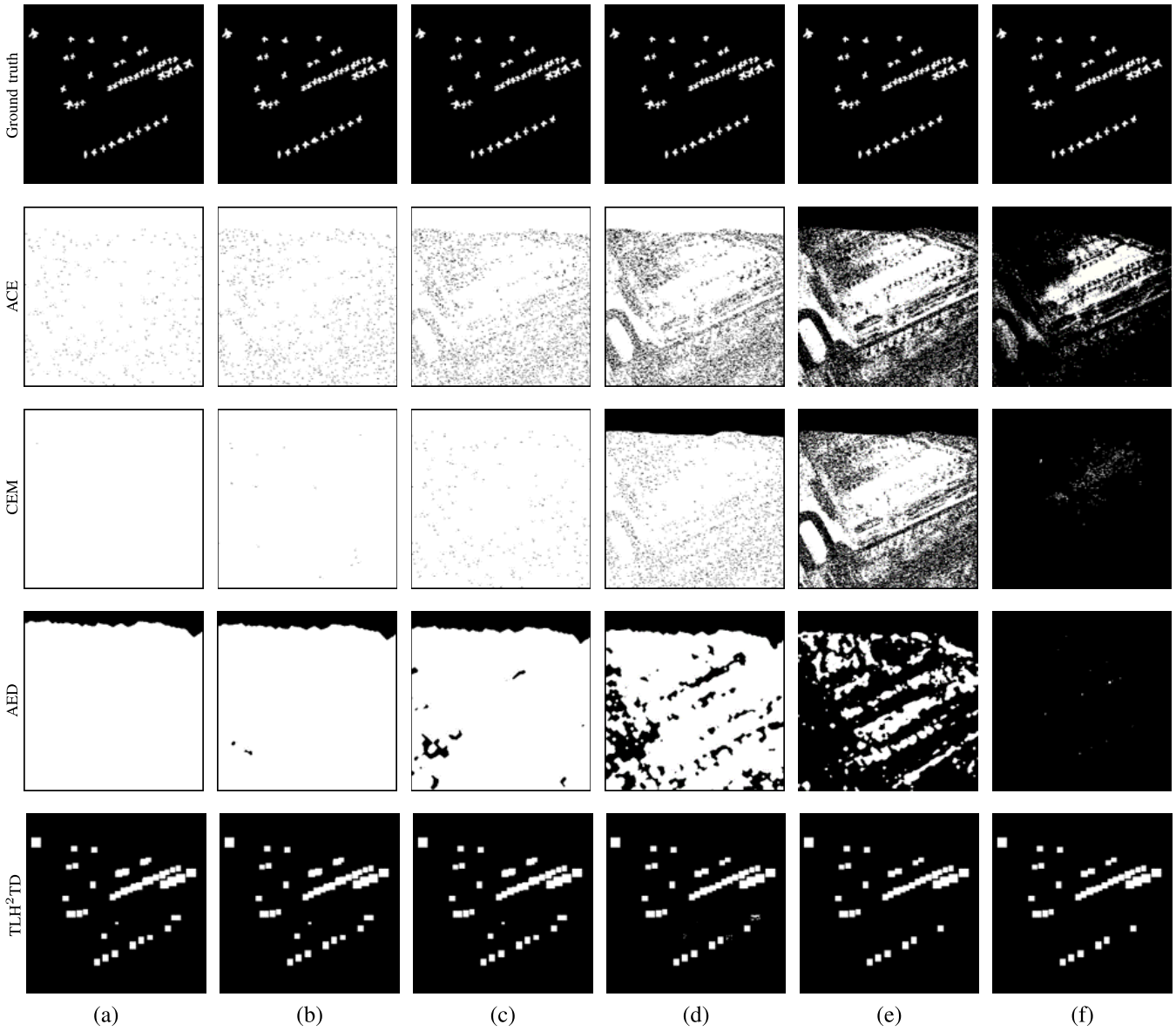



Fig. 27. Comparison of binarization results of different HTD algorithms for the L-f06 dataset: (a) thre = 1e-6, (b) thre = 1e-5, (c) thre = 1e-4, (d) thre = 1e-3, (e) thre = 1e-2, and (f) thre = 0.1. Note that thre denotes the threshold for target and background segmentation.

advanced HTD algorithms. Table IV lists the AUC values and time cost for detection.

Figs. 20–22 show the comparison of the target detection results and ROC curves for the two data.

In Table IV, the column "Required a priori knowledge" represents whether the algorithm requires spatial or spectral prior knowledge, and the column "Available" represents whether the algorithm works on these HSIs. Especially, in the column "Available," "Out of memory" means that during the calculations, the complexity of the HTD algorithm's parameters leads to excessive resource usage during the calculation, exceeding the server's memory limits. "Array over limit" means that due to the complex design of the corresponding HTD algorithm, it usually carries out complex calculations by constructing large arrays during the calculations, which results in the length of the arrays being too large and cannot be achieved on servers with limited computing resources as well. In both cases, the corresponding algorithms cannot meet the needs of onboard real-time high-precision HTD with limited computing resources. In Table IV, the AUC values and ROC demonstrate that the proposed TLH$^2$TD method has the best detection performance, and the average AUC value is 0.9566, representing accuracy improvements over the other algorithms of 26.5%. Five out of 19 algorithms failed because the array was out of limit or memory was exceeded, demonstrating that these algorithms cannot meet the needs of onboard real-time processing. Among the remaining 14 algorithms, ten algorithms have average AUC values below 0.8, indicating that they had failed on the medium-sized HSIs. The LocalKernel-RX, SSIIFD, and GTVLRR algorithms, which have average AUC values higher than 0.8, take hundreds or even thousands of seconds. Specifically, SSIIFD and KIFD, which perform well on small-sized HSIs, have low performance on medium-sized HSIs. The SSIIFD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

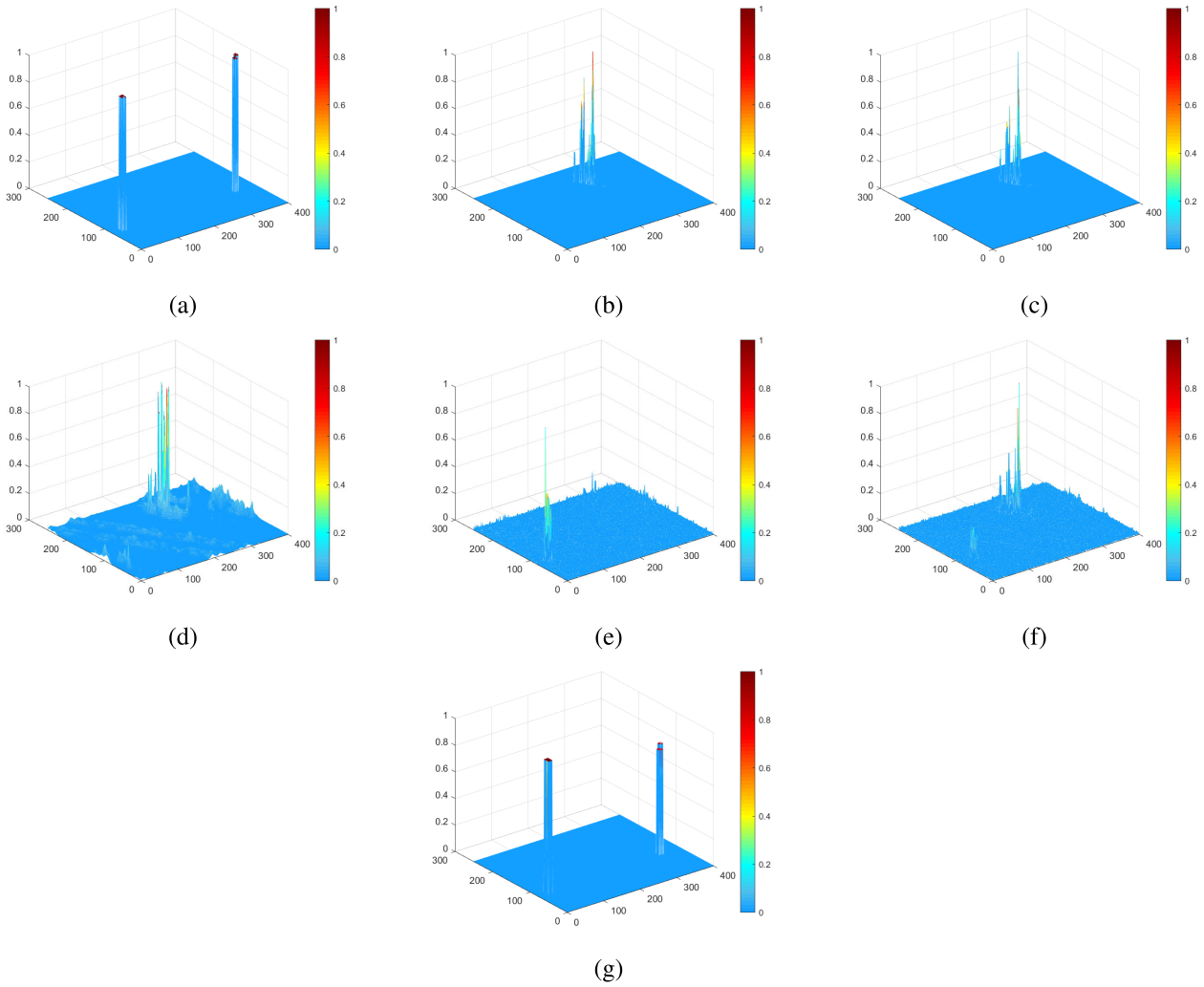



Fig. 28. 3-D plots of L-f08 data for different algorithms after normalizing the detection results (the areas shown are cropped from the original image and contain aircraft): (a) ground truth, (b) RXD, (c) FrFE, (d) AED, (e) ACE, (f) CEM, and (g) TLH$^2$TD.

algorithm takes too long and has an average AUC value of only 0.8024, and the KIFD algorithm failed due to a lack of memory. This further demonstrates the effectiveness and robustness of TLH$^2$TD for data processing in complex scenarios.

The average detection time of TLH$^2$TD is faster than that of other algorithms and slightly longer than that of RXD (0.165 s). The detection results and the ROCs show that the proposed method enhances the airplane targets and suppresses the background, demonstrating good detection performance.

*3) Results of TLH$^2$TD and Other HTD Algorithms for Large-Sized HSIs:* Table V lists the comparison of the AUC values and the time cost of the TLH$^2$TD and other advanced HTD algorithms on the L-f08 and L-f06 datasets. The number of bands in L-f06 was downscaled due to the high data dimensionality, and 112 bands were selected for testing.

Figs. 23–25 show the comparison of the target detection results and ROC curves for L-f08 and L-f06.

Table V indicates that 13 out of 19 HTD algorithms failed the target detection for large images, such as local-RX, kernel-RX, LRASR, LSMAD, and other algorithms. The reasons were large matrix dimensions of intermediate results, exceeding the maximum array limit, timeout, or insufficient memory. For LocalKernel-RX and GTVLRR, the time costs more than 9 h, indicating that the two algorithms failed on large-sized HSIs. Although the FrFE algorithm obtains good results, the operation time is too long; thus, it does not meet the needs of real-time high-precision target detection. The RXD algorithm is unstable, with an average AUC value of 0.7984 despite its low time consumption. Thus, it does not meet the needs of robust HTD. The AUC value and algorithm time efficiency of the AED and ACE algorithms are worse than those of the TLH$^2$TD algorithm for both datasets. Only the CEM algorithm is slightly more time-efficient than the TLH$^2$TD and has a slightly lower AUC value. However, the detection result on the L-f08 dataset shows that the TLH$^2$TD algorithm can detect high-visibility and dark aircraft, whereas the CEM algorithm only detects high-visibility aircraft. On the L-f06 dataset, the TLH$^2$TD algorithm can distinguish the target from the background, whereas the CEM algorithm shows worse performance.

Meanwhile, as can be seen from Tables IV and V, the proposed TLH$^2$TD has potential for onboard real-time processing. The main reasons include the following.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

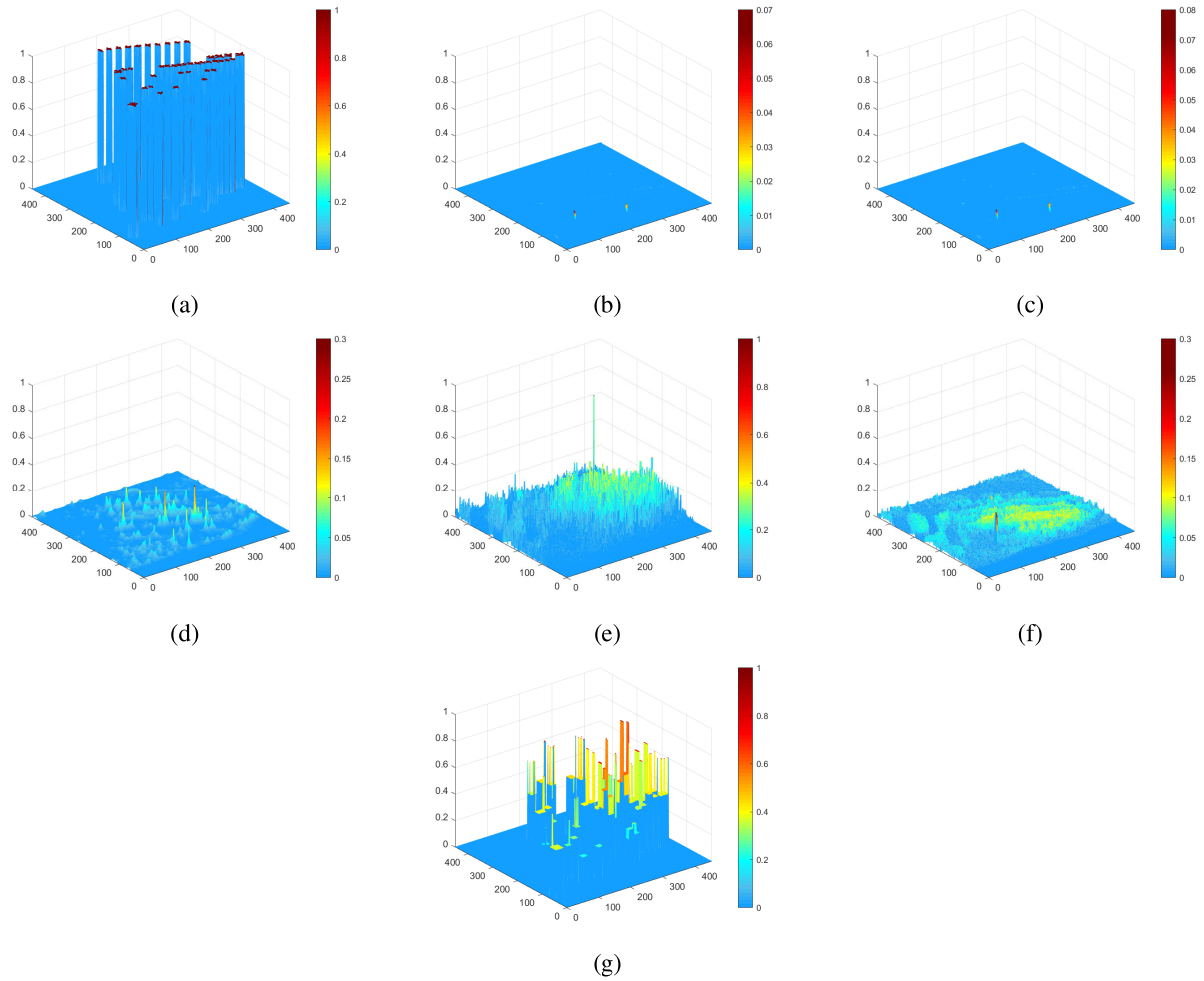



Fig. 29. 3-D plots of L-f06 data for different algorithms after normalizing the detection results (the areas shown are cropped from the original image and contain aircraft): (a) ground truth, (b) RXD, (c) FrFE, (d) AED, (e) ACE, (f) CEM, and (g) TLH$^2$TD.

1) Onboard hyperspectral data processing is mostly oriented toward raw uncropped whole data. Consistent with onboard data processing, the proposed algorithm is mainly designed for the processing of large-scale HSIs, while other methods are mostly oriented to small-sized standard data.
2) The TLH$^2$TD algorithm is specifically designed to mitigate the necessity of data stacking during onboard real-time processing. Instead of using complex computational processes or oversized matrices, the TLH$^2$TD algorithm uses an effective weighted fusion of spatial–spectral features to achieve improved detection accuracy and algorithmic efficiency.
3) According to the comparison of the time cost of different HTD algorithms in Tables IV and V, it can be seen that most HTD algorithms have already failed due to insufficient memory or excessive time consumption on medium- and large-sized HSIs, whereas the proposed TLH$^2$TD costs less time while achieving high accuracy, demonstrating its potential for onboard real-time processing.

In addition, the AUC value is pixel-based. In real-time processing, the primary goal is target detection and background suppression, and it is not required to detect all pixels of each target. The ROC curve takes the false alarm rate and detection rate under different thresholds as the $x$-axis and the $y$-axis, respectively, and the AUC value is the area under the ROC curve. From the calculation process of AUC, it can be seen that AUC is not sensitive to imbalances between positive

TABLE VI
ABLATION EXPERIMENT WITH/WITHOUT THE CTBSE MODULE

| HTD Method | No CTBSE | CTBSE | Average AUC of HSIs |
|---|---|---|---|
| TLH$^2$TD | √ | | 0.8881 |
| TLH$^2$TD | | √ | **0.9607** |

TABLE VII
ABLATION EXPERIMENT FOR DIFFERENT DATA AUGMENTATION METHODS

| Guassian filter $\alpha$ | Rotation angle | Scale | Ratio of images with pasted airplanes | Average AUC of HSIs |
|---|---|---|---|---|
| [0.2,0.4,0.6,0.8,1.0] | 0-360° | [0.8,1.2] | 100% | 0.9472 |
| [0.8,0.9,1.0] | 0-360° | [0.8,1.2] | 100% | 0.9554 |
| [0.8,0.9,1.0] | 0-360° | [0.8,1.2] | 50% | 0.9524 |
| [0.6,0.7,0.8,0.9,1.0] | 0-360° | [0.8,1.2] | 50% | **0.9607** |
| [0.6,0.7,0.8,0.9,1.0] | 0-360° | [0.7,1.3] | 50% | 0.9524 |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

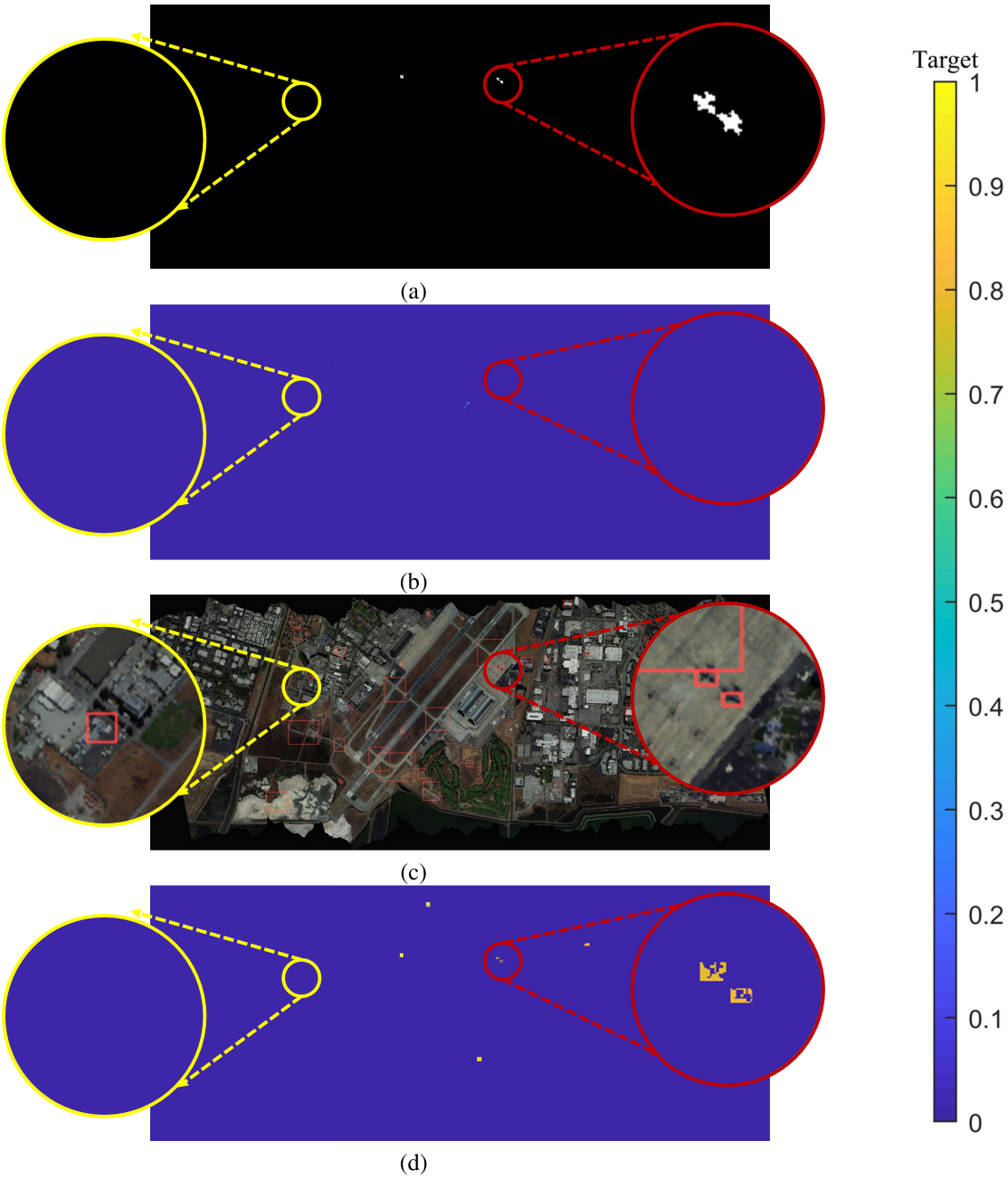



Fig. 30. Detection results of L-f08 by different features: (a) ground truth, (b) detection result by only spectral features (RXD algorithm), (c) detection result by only spatial features (Yolo V5), and (d) detection result by fused spectral–spatial features (proposed TLH$^2$TD). The red circle represents a false positive in RXD, and the yellow circle represents a false negative in Yolo V5, while the proposed TLH$^2$TD can correctly distinguish them.

and negative samples. As long as the variation relationship between the false alarm rate and the detection rate is close, the AUC values of different HTD algorithms will be close, which cannot accurately assess the algorithms' effectiveness in large-scale HSIs with imbalanced samples. Despite the high AUC values of other algorithms, such as the CEM, they are not effective in distinguishing between the target and the background in practice. Thus, it is difficult to evaluate the algorithm performance for target detection in large-scene HSIs by relying solely on the ROC and AUC values. Therefore, we selected the classification thresholds 1e-6, 1e-5, 1e-4, 1e-3, 1e-2, and 1e-1 in probability maps of the algorithms to obtain binary target-background images to verify the algorithm's performance in target detection and background suppression, as shown in Figs. 26 and 27. In addition, Figs. 28 and 29 show the 3-D plots after normalizing the detection results. The plots are cutouts with airplanes cropped from the images to compare the aircraft detection results.

For smaller HSIs, the effectiveness of HTD algorithms can be assessed with AUC values. However, for large-scale HSIs with complex scenes and imbalanced samples, this evaluation metric is rather difficult to accurately evaluate the performance of the algorithms. As shown in Figs. 26 and 27, at very low thresholds (e.g., when thre = 1e-5), the TLH$^2$TD method can distinguish the target from the background accurately, demonstrating that the background suppression performance is excellent. Although the ACE, AED, and CEM algorithms have higher AUC values than the TLH$^2$TD, the false alarm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

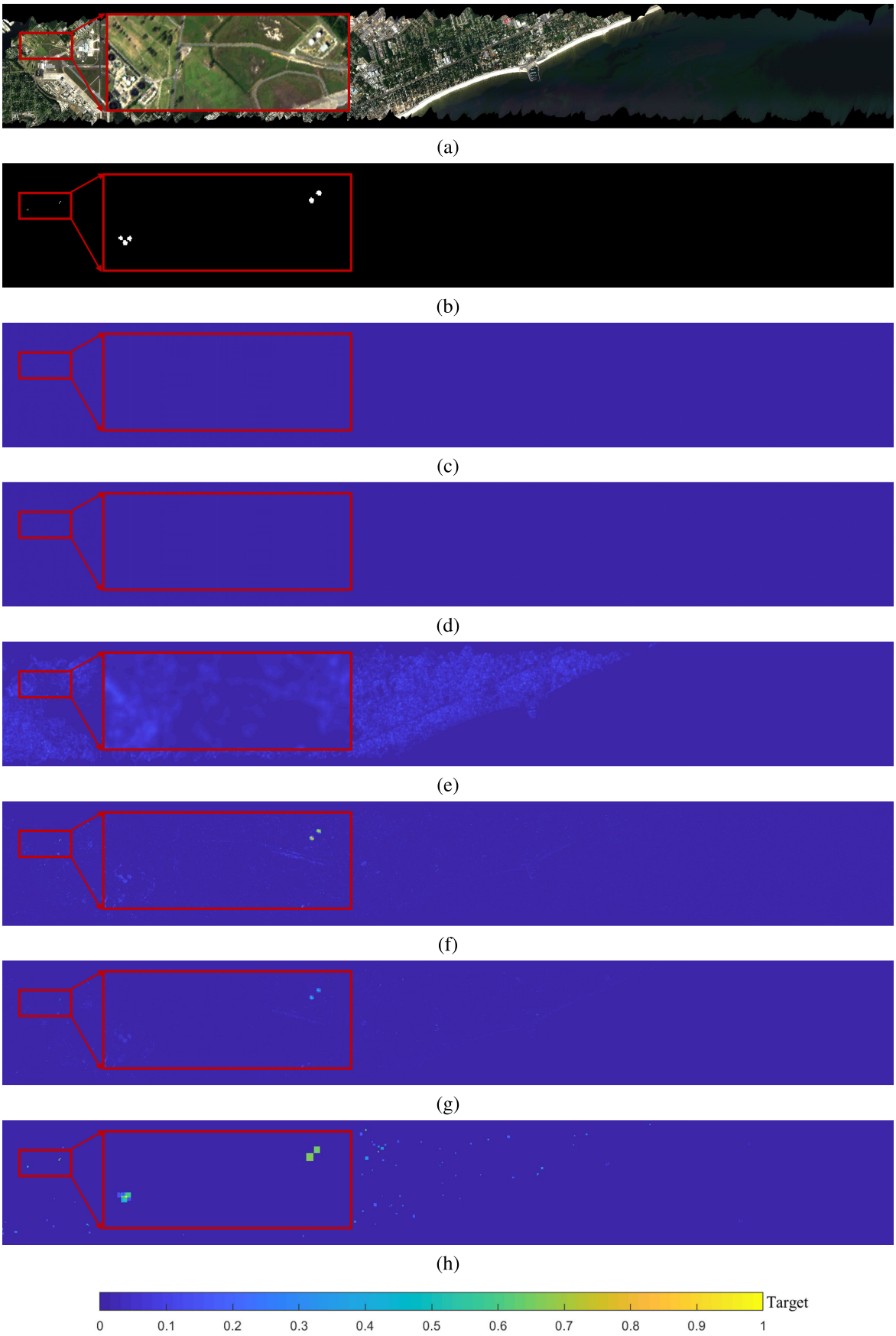



Fig. 31. Detection results of different algorithms for the L-f11 dataset: (a) true color image, (b) ground truth, (c) RXD, (d) FrFE, (e) AED, (f) ACE, (g) CEM, and (h) TLH$^2$TD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

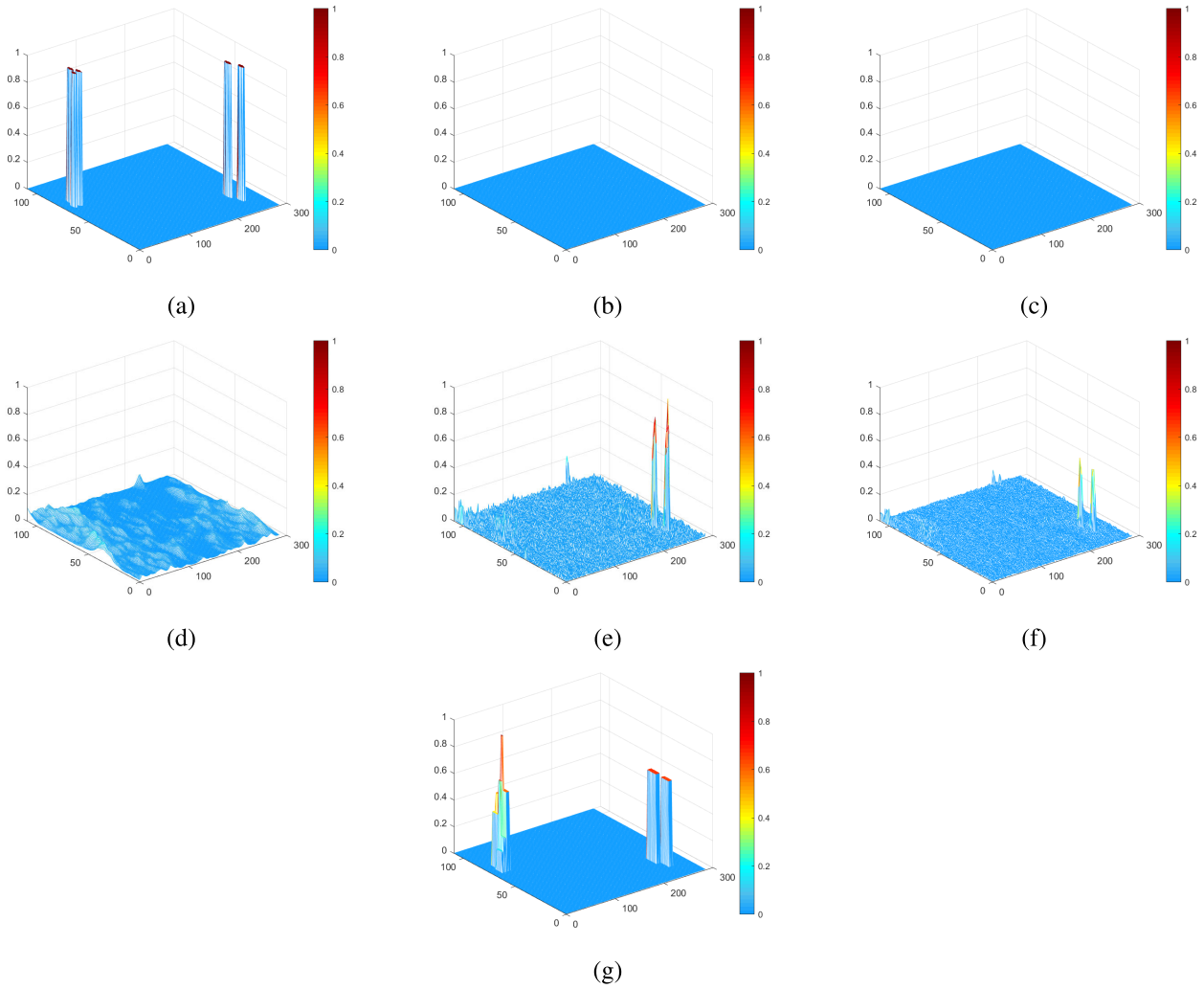



Fig. 32. 3-D plots of L-f11 data for different algorithms after normalizing the detection results (the areas shown are cropped from the original image and contain storage tanks): (a) ground truth, (b) RXD, (c) FrFE, (d) AED, (e) ACE, (f) CEM, and (g) TLH$^2$TD.

rate is very high for large remote sensing images. A better result is obtained by the three algorithms for bright airplanes in the L-f08 dataset when the threshold is 0.1, but the two dark airplanes on the right side are not detected. However, the TLH$^2$TD method successfully detects these three airplanes. For the L-f06 dataset, the ACE, AED, and CEM algorithms cannot adequately distinguish the target and background when the threshold is 0.1. The aircraft is black in the binarization result of the ACE algorithm, i.e., it has been identified as the background, and the background has been classified as the target. The CEM algorithm confuses the target and the background, indicating that it is unsuitable for practical detections, although the AUC value is relatively high, and the algorithm is highly efficient. The TLH$^2$TD algorithm has a high AUC value and exhibits superior target-background separation, making it more suitable for robust HTD in large scenes.

Figs. 28 and 29 indicate that the RXD and FrFE algorithms had low detection performance. Although the AUC values of the two algorithms are relatively high, both the target and background have low values, making it difficult to differentiate them. The AED, ACE, and CEM algorithms cannot distinguish the target from the background utilizing a single threshold, and the false alarm rate is high. In contrast, the TLH$^2$TD algorithm utilizes a larger portion of the threshold range to distinguish the targets from the background. In summary, the proposed TLH$^2$TD algorithm is more robust than other algorithms in detecting targets in large-scene HSIs and provides higher detection efficiency and accuracy. Thus, it can be used in real time for onboard HSI processing.

### D. Ablation Study

The following ablation experiments were conducted.

1) *CTBSE Module:* We compared the average AUC of HSIs of the HTD network with and without the CTBSE module. As shown in Table VI, the average AUC is 8.17% higher with the CTBSE module, demonstrating the validity of data transformation.
2) *Different Data Augmentation Parameters of Copy–Paste and Mixup:* We compare the effects of different data augmentation parameters of copy–paste and Mixup, including the Gaussian filter $\alpha$, the rotation angle and scaling of the pasted samples, and the ratio of RGB images with pasted airplanes in the CTBSE module. As shown in Table VII, the detection accuracy of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
COMPARISON OF AUC VALUES AND TIME COST FOR TLH$^2$TD AND OTHER HTD ALGORITHMS FOR LARGE-SIZED HSIS WITH STORAGE TANKS

| Algorithms | Required a priori knowledge | Available | AUC | | | | Time Cost (second) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | L-f06 | L-f09 | L-f11 | Average | L-f06 | L-f09 | L-f11 |
| RXD | × | √ | 0.8835 | 0.3405 | 0.4071 | 0.5437 | 31.04 | 28.29 | 33.21 |
| Local-RX | × | Array over limit | / | / | / | / | / | / | / |
| FrFE | × | √ | 0.9127 | 0.5336 | 0.6006 | 0.6823 | 4507.64 | 3321.31 | 4209.43 |
| Kernel-RX | × | Array over limit | / | / | / | / | / | / | / |
| LocalKernel-RX | × | Timeout(>9h) | / | / | / | / | / | / | / |
| RPCA | × | Array over limit | / | / | / | / | / | / | / |
| LRASR | × | Out of memory | / | / | / | / | / | / | / |
| LSMAD | × | Out of memory | / | / | / | / | / | / | / |
| SSIIFD | × | Out of memory | / | / | / | / | / | / | / |
| PTA | × | Out of memory | / | / | / | / | / | / | / |
| GTVLRR | Spatial | Timeout(>9h) | / | / | / | / | / | / | / |
| KIFD | Spatial | Out of memory | / | / | / | / | / | / | / |
| GAE | Spatial | Out of memory | / | / | / | / | / | / | / |
| Auto-AD | Spatial | Out of memory | / | / | / | / | / | / | / |
| RGAE | Spatial | Out of memory | / | / | / | / | / | / | / |
| AED | Spatial | √ | 0.3675 | 0.6485 | 0.7513 | 0.5891 | 86.10 | 78.78 | 82.20 |
| ACE | Spectral | √ | 0.2685 | 0.7216 | 0.9710 | 0.6537 | 53.85 | 43.37 | 51.94 |
| CEM | Spectral | √ | 0.7593 | 0.8604 | **0.9815** | 0.8671 | 25.02 | 20.62 | 25.83 |
| TLH$^2$TD | Spatial | √ | **0.9368** | **0.9979** | 0.9799 | **0.9715** | 31.73 | 28.95 | 33.96 |

network is the highest with a Gaussian filter $\alpha$ of [0.6, 0.7, 0.8, 0.9, 1.0], a rotation angle of 0°–360°, a scale of [0.8, 1.2], and 50% images pasted with airplanes in the CTBSE module.

3) *Spatial–Spectral Feature Fusion Module:* We compared the target detection results of L-f08 using only spectral features (RXD algorithm), only spatial features (Yolo v5x), and fused spectral–spatial features (proposed TLH$^2$TD). As shown in Fig. 30, the detection results of RXD and Yolo V5 show some missing targets and false alarms, such as the aircraft target missed in the RXD result (red circle) and the background pixel detected as a target in the Yolo v5x result (yellow circle). In contrast, the targets and background are correctly detected by the proposed TLH$^2$TD. Therefore, our proposed TLH$^2$TD can substantially reduce the false alarm rate and improve the separability of targets and the background for large-scale HSIs.

4) *Experiments on Large-Scale HSIs With Storage Tanks as Targets:* We compared the target detection results of three HSIs L-f06 (6122 × 922 × 224, 2.1-m resolution), L-f09 (6144 × 747 × 224, 16.8-m resolution), and L-f11 (6122 × 854 × 224, 3.3-m resolution) with storage tanks as targets. As shown in Table VIII, 13 out of 19 HTD algorithms failed, and TLH$^2$TD has the highest average AUC value of 0.9715 among the remaining algorithms. From Figs. 31 and 32, it can be seen that TLH$^2$TD can better distinguish the storage tanks from background. Although the AUC value of the CEM algorithm on L-f11 is slightly higher than TLH$^2$TD, it can be seen from the 3-D plots that only two out of five storage tanks are detected by CEM, whereas our method detects five tanks and suppresses the background well. The results on HSIs with storage tanks proved the generalizability and robustness of TLH$^2$TD across various scenarios, further demonstrating its effectiveness.

## V. CONCLUSION

This article proposed a robust HTD algorithm for aircraft detection in large-scene HSIs called TLH$^2$TD based on transfer learning and spatial–spectral feature fusion. More specifically, TLH$^2$TD expands hyperspectral labeled samples by the CTBSE module with a few hyperspectral labeled data and abundant high-resolution RGB labeled images. In addition, the spatial features of hyperspectral targets are fully extracted by the HPNEM module in the network training phase. Then, the SSWS module is designed to fuse spatial and spectral features to achieve high-precision target detection. It addresses the problems of insufficient utilization of spatial features, scarce training samples, and inadequate target characterization in large-scale HSIs by existing HTD methods. The algorithm detects the target and suppresses the background more accurately and efficiently than existing hyperspectral AED algorithms. In addition, experiments on large-scale HSIs demonstrate the accurate and robust detection performance of the TLH$^2$TD algorithm, indicating its potential for onboard real-time processing. In the future, we plan to study the simultaneous extraction of spatial and spectral features from HSIs to improve the algorithm's efficiency and portability.

## REFERENCES

[1] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, Jun. 1985.

[2] B. Zhang, "Advancement of hyperspectral image processing and information extraction," *J. Remote Sens.*, vol. 20, no. 5, pp. 1062–1090, 1985.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

[3] Q. Tong, B. Zhang, and L. F. Zheng, *Hyperspectral Remote Sensing: Principle, Technology and Application*. Beijing, China: Higher Educ. Press, 2006.

[4] L. P. Zhang and H. F. Shen, "Progress and future of remote sensing data fusion," *J. Remote Sens.*, vol. 20, no. 5, pp. 1050–1061, 2006.

[5] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[6] M. T. Eismann, *Hyperspectral Remote Sensing*. Bellingham, WA, USA: SPIE, 2012.

[7] B. Zhang and L. R. Gao, *Hyperspectral Image Classification Target Detection*. Beijing, China: Sci. Press, 2011.

[8] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[9] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[10] E. A. Ashton and A. Schaum, "Algorithms for the detection of sub-pixel targets in multispectral imagery," *Photogramm. Eng. Remote Sens.*, vol. 64, no. 7, pp. 723–731, 1998.

[11] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[12] Y. P. Taitano, B. A. Geier, and K. W. Bauer, "A locally adaptable iterative RX detector," *EURASIP J. Adv. Signal Process.*, vol. 2010, no. 1, pp. 1–12, May 2010.

[13] D. Liu and L. Han, "Spectral curve shape matching using derivatives in hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 504–508, Apr. 2017.

[14] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[15] W.-S. Chen and I. S. Reed, "A new CFAR detection test for radar," *Digit. Signal Process.*, vol. 1, no. 4, pp. 198–214, Oct. 1991, doi: 10.1016/1051-2004(91)90113-y.

[16] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.

[17] J. C. Harsanyi, *Detection and Classification of Subpixel Spectral Signatures*. College Park, MD, USA: Univ. Maryland, 1993.

[18] H. Ren and C.-I. Chang, "A target-constrained interference-minimized filter for subpixel target detection in hyperspectral imagery," in *Proc. IGARSS*, vol. 4, Nov. 2000, pp. 1545–1547.

[19] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[20] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2017.

[21] C. Zhao, C. Li, S. Feng, and N. Su, "Spectral–spatial stacked autoencoders based on the bilateral filter for hyperspectral anomaly detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2209–2212.

[22] S. Feng, D. Wu, R. Feng, and C. Zhao, "Hyperspectral anomaly detection with total variation regularized low rank tensor decomposition and collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] J. Du, Z. Li, and H. Sun, "CNN-based target detection in hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 2761–2764.

[24] W. Li, G. Wu, and Q. Du, "Transferred deep learning for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5177–5180.

[25] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020, doi: 10.3390/rs12091489.

[26] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[27] Z. Zhao and B. Sun, "Hyperspectral anomaly detection via memory-augmented autoencoders," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1274–1287, Jun. 2022, doi: 10.1049/cit2.12116.

[28] J. Wei, J. Zhang, Y. Xu, L. Xu, Z. Wu, and Z. Wei, "Hyperspectral anomaly detection based on graph regularized variational autoencoder," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[29] X. Wang, L. Wang, and Q. Wang, "Local spatial–spectral information-integrated semisupervised two-stream network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535515.

[30] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[31] F. Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jan. 2021.

[32] M. Iman, H. R. Arabnia, and K. Rasheed, "A review of deep transfer learning and recent advancements," *Technologies*, vol. 11, no. 2, p. 40, Mar. 2023.

[33] K. Weiss, T. M. Khoshgoftaar, and D. D. Wang, "A survey of transfer learning," *J. Big Data*, vol. 3, no. 1, pp. 1–40, 2016.

[34] G. Jun and J. Ghosh, "An efficient active learning algorithm with knowledge transfer for hyperspectral data analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2008, pp. 52–55.

[35] B. Du, L. Zhang, D. Tao, and D. Zhang, "Unsupervised transfer learning for target detection from hyperspectral images," *Neurocomputing*, vol. 120, pp. 72–82, Nov. 2013.

[36] J. Shen, X. Cao, Y. Li, and D. Xu, "Feature adaptation and augmentation for cross-scene hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 622–626, Apr. 2018.

[37] L. Bao and Y. Zhu, "Research on hyperspectral image target detection by the convergence neural network based on transfer learning," in *Proc. IEEE Int. Conf. Prog. Informat. Comput. (PIC)*, Dec. 2018, pp. 95–100.

[38] X. Li, L. Zhang, B. Du, L. Zhang, and Q. Shi, "Iterative reweighting heterogeneous transfer learning framework for supervised remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 2022–2035, May 2017.

[39] Y. Shi, J. Li, Y. Li, and Q. Du, "Sensor-independent hyperspectral target detection with semisupervised domain adaptive few-shot learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6894–6906, Aug. 2021.

[40] J. Lei, S. Xu, W. Xie, J. Zhang, Y. Li, and Q. Du, "A semantic transferred priori for hyperspectral target detection with spatial–spectral association," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507314.

[41] Z. Huang, X. Kang, S. Li, and Q. Hao, "Game theory-based hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2965–2976, Apr. 2020.

[42] B. Tu, Z. Wang, X. Yang, J. Li, and A. Plaza, "Hyperspectral anomaly detection using quantum potential clustering," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.

[43] X. Song, S. Aryal, K. M. Ting, Z. Liu, and B. He, "Spectral–spatial anomaly detection of hyperspectral data based on improved isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516016.

[44] G. Ghiasi et al., "Simple copy-paste is a strong data augmentation method for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2918–2928.

[45] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[46] A. Shrivastava, A. Gupta, and R. Girshick, "Training region-based object detectors with online hard example mining," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 761–769.

[47] J. Shermeyer, T. Hossler, A. V. Etten, D. Hogan, R. Lewis, and D. Kim, "RarePlanes: Synthetic data takes flight," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 207–217.

[48] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[49] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1–37, May 2011, doi: 10.1145/1970392.1970395.

[50] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[51] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.

[52] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.
[53] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.
[54] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.
[55] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.
[56] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.
[57] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.



**Yuanfeng Wu** (Senior Member, IEEE) received the B.S. and M.S. degrees in computer science from the China University of Mining and Technology, Beijing, China, in 2004 and 2007, respectively, and the Ph.D. degree in cartography and geographical information systems from the Graduate University of Chinese Academy of Sciences, Beijing, in 2010.

He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include the development of onboard real-time algorithms, high-performance computing implementation, and computer software in hyperspectral image processing.



**Zijin Li** received the B.S. degree in geographic information science from Peking University, Beijing, China, in 2021. She is currently pursuing the M.S. degree in signal and information processing with the Key Laboratory of Computational Optical Imaging Technology, School of Electronic, Electrical and Communication Engineering, Aerospace Information Research Institute, Chinese Academy of Sciences, University of Chinese Academy of Sciences, Beijing.

Her research interests include hyperspectral image processing and object detection.



**Boya Zhao** was born in 1990. He received the B.Sc. degree from the School of Electrical Engineering and Information, Hebei University of Technology, Tianjin, China, in 2013, and the Ph.D. degree from the School of Electrical and Information Engineering, Beijing Institute of Technology, Beijing, China, in 2019.

He is currently an Assistant Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include object detection in complex background and onboard real-time information processing.



**Yuhang Song** received the B.E. degree in spatial information and digital technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2023. He is currently pursuing the Ph.D. degree in cartography and geographical information systems with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China.

His research interests include remote sensing image processing, deep learning, and oriented object detection.



**Bing Zhang** (Fellow, IEEE) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS, where he has been leading lots of key scientific projects in the area of hyperspectral remote sensing for more than 25 years. He is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing. He has developed five software systems in image processing and applications. He has authored more than 300 publications, including more than 200 journal articles. He has edited eight books/contributed book chapters on hyperspectral image processing and subsequent applications. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang has been a Technical Committee Member of the IEEE Workshop on Hyperspectral Image and Signal Processing since 2011 and the President of the Hyperspectral Remote Sensing Committee of the China National Committee of the International Society for Digital Earth since 2012. He got the IEEE GRSS Regional Leader Award in 2020. He is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:27:41 UTC from IEEE Xplore. Restrictions apply.